DC-GUARDIAN Group Project Report

Intelligent Reasoning Systems Project · Project title: *DC-GUARDIAN: Multi-Domain Reasoning for Data-Center Security, Safety and Operations*

Practice Module Group 22 · Graduate Certificate in Intelligent Reasoning Systems, NUS-ISS

Member	Student ID
AL AJMI ALJAWHARAH MOHAMMAD H	A0356913X
ALOTAIBI MARAM MUGAAD M	A0355452A
ALSHAKHS MAI ZAID M	A0354315J
ALSHAMMARY NORAH MOHAMMED H	A0354356Y

Contents

DC-GUARDIAN Group Project Report 1
1. Introduction 6
2. Project Background / Market Context 7
2.1 Market landscape 10
3. Literature Review / Market Research 10
3.1 Literature review 10
3.2 Market research: existing solutions 11
3.3 Business case 13
4. Project Scope 13
4.1 Objectives and how each was measured 14
4.2 Intelligent Reasoning 14
5. Data Collection and Preparation 15
5.1 Data and knowledge sources 16
5.2 Preparation 17
5.3 Labelled scenarios 17
6. System Design 18
6.1 Design requirements 18
6.2 Architecture overview 20
6.3 Layer-by-layer design 22
6.4 Domain model and Common Event Schema 24
6.5 Choosing the reasoning techniques 26
6.6 Worked example: the Zone A incident 29
6.7 Design principles 30
6.8 Changes from the proposal 30
7. Implementation 33
7.1 Evidence layer 33
7.2 Reasoning layer 37
7.3 Response layer 40
7.4 Decision layer 42
7.5 Presentation layer 44
7.6 Testing approach 49
7.7 Technical challenges and how they were resolved 50
8. Results and Progress 50
8.1 Detector evaluations (locked test sets, run once) 50

8.2 Correlation against isolated monitoring (research question) 52
8.3 Bayesian correlation against the crisp rule 53
8.4 Severity rules 53
8.5 Knowledge grounding and language-model safeguards 53
8.6 Latency 53
8.7 Usability study 53
8.8 Deviations from the initial plan 56
8.9 Experiment register 56
9. Challenges 59
9.1 Project Challenges 59
9.2 Strategy for Improvement 61
10. Future Work 63
11. Conclusion 64
Supplementary Materials 65
Appendix A. Project Proposal 65
Appendix B. Mapping of System Functions to Course Techniques 66
Appendix C. Installation and User Guide 66
C.1 Requirements 66
C.2 Installation 67
C.3 Running the system 67
C.4 Verifying the installation 67
C.5 Using the dashboard 67
C.6 Data not included 68
Appendix D. Individual Project Reports 68
D.1 AL AJMI ALJAWHARAH MOHAMMAD H (A0356913X) 68
D.2 ALOTAIBI MARAM MUGAAD M (A0355452A) 68
D.3 ALSHAKHS MAI ZAID M (A0354315J) 69
D.4 ALSHAMMARY NORAH MOHAMMED H (A0354356Y) 70
Appendix E. Generative AI Use Disclosure 70
Appendix F. References 71
Abbreviations 73
Acknowledgements 74

List of Figures

Figure 1- System overview: five evidence sources, the DC-01 knowledge graph and the intelligence core 20
Figure 1- System overview: five evidence sources, the DC-01 knowledge graph and the intelligence core 20
Figure 2- DC-GUARDIAN architecture: five layers and the human review loop 21
Figure 2- DC-GUARDIAN architecture: five layers and the human review loop 21
Figure 3 - Detailed DC-GUARDIAN evidence-processing workflow from detector output to human review 22
Figure 4 - Overview of Neo4j graph 38
Figure 5. Monitoring Center: summary cards, domain counts ("N abnormal of M"), the help panel and the event feed, with the HIGH · LINKED Zone A group selected 45
Figure 6. Group detail: the At a glance summary (top) and the Neo4j graph linked to the zone (bottom); Full detail also shows the Decision rules, events and verdicts, and the investigation state 47
Figure 7. Human Review Queue: a recorded verdict, the audit-chain confirmation and the verdict history 47
Figure 8. AI Investigator: the operator's question 48
Figure 9. AI Investigator: the full answer, with its claim checks and sources 48

List of Tables

Table 1. Existing solutions and how DC-GUARDIAN differs 12
Table 2. Indicative adoption tiers 13
Table 3. Objectives, success measures and outcomes 14
Table 4. Data and knowledge sources 16
Table 5. Design requirements and how the design meets them 19
Table 6. Each layer's purpose, inputs, outputs and limits 22
Table 7. Node types in the DC-01 knowledge graph 24
Table 8. Relationships in the DC-01 knowledge graph 25
Table 9. Common Event Schema fields 25
Table 10. Options compared for deciding which events belong together 27
Table 11. One technique per job 29
Table 12. The Zone A incident, layer by layer 29
Table 13. Changes from the original proposal 31
Table 14. The five evidence detectors 33
Table 15. SSH behavioral window features 34
Table 16. SMART attributes used by the drive model 35
Table 17. Deterministic correlation rules 38
Table 18. Bayesian network factors and their states 39
Table 19. Expert conditional probabilities (selected states) 39
Table 20. Knowledge-base documents (13 approved; candidates and the planned policy marked) 40
Table 21. Specialist agents and their domains 42
Table 22. DCG-DECISION-v1 rules 43
Table 23. DCG-DECISION-v2 single-event severities 43
Table 24. Candidate group linkage rules 44
Table 25. Roles and what each can see 49
Table 26. Technical challenges and how they were resolved 50

Table 27. Detector evaluations on locked test sets 50
Table 28. SSH detector on locked Set B 51
Table 29. Drive failure risk on the 2026 Q1 holdout 51
Table 30. PPE detectors on the SH17 locked test 52
Table 31. Correlation against isolated monitoring (test split) 52
Table 32. Bayesian correlation against the crisp rule 53
Table 33. Per-stage latency 53
Table 34. Usability study participants and SUS scores 54
Table 35. Usability task results 54
Table 36. Concept check: what participants understood 55
Table 37. Trust and clarity ratings 55
Table 38. Deviations in what was delivered and evaluated 56
Table 39. Experiment register 57
Table 40. Model and evaluation limitations by component 59
Table 41. Remaining challenges and strategies 60
Table 42. Improvement actions by component 62
Table 43. Challenge strategy summary (C1 to C7) 62
Table 44. Proposal commitments and their status 65
Table 45. Mapping of system functions to course techniques 66
Table 46. Generative AI use 70

1. Introduction

Data centers are critical infrastructures that support essential digital services, business operations, and information systems. Maintaining their security, safety, and operational reliability requires continuous monitoring of physical security, cybersecurity threats, equipment health, and environmental conditions that may affect operational performance. However, these aspects are often managed through separate tools and systems, making it difficult to identify relationships between events and recognize potential risks at an early stage. For example, an unauthorized individual entering a data center without the required Personal Protective Equipment (PPE), followed by repeated unauthorized SSH login attempts, may generate several unrelated alerts despite potentially being part of a broader security incident. Similarly, abnormal environmental conditions, such as excessive temperature or humidity, may contribute to equipment degradation and eventual hardware failure without being recognized as connected events. The consequences of failing to identify and respond to such incidents can be significant. According to the 2024 survey reported in Uptime Institute's Annual Outage Analysis 2025, 54% of operators indicated that their most recent significant outage cost more than US$100,000, while one in five reported costs exceeding US$1 million [1].

DC-GUARDIAN is an intelligent, unified data center protection and maintenance monitoring system designed to integrate information from five monitoring domains into a single interactive dashboard. These domains include physical access control, personnel safety, cybersecurity, hardware health, and environmental monitoring. The system incorporates facial recognition to identify authorized and unauthorized personnel, PPE detection to monitor safety compliance, SSH anomaly detection to identify suspicious cybersecurity activities, hard-drive failure risk prediction to support proactive maintenance, and environmental threshold monitoring to detect abnormal conditions such as temperature and humidity. By integrating these capabilities, DC-GUARDIAN aims to move beyond isolated event detection toward intelligent cross-domain incident correlation, providing operators with a more comprehensive understanding of potential security, safety, and maintenance risks.

The system is developed using a five-layer architecture that integrates Artificial Intelligence (AI), sensor monitoring, knowledge graphs, Bayesian reasoning, and Retrieval-Augmented Generation (RAG). The Evidence Layer collects information from five specialized detectors, each responsible for identifying events within its monitoring domain. The Reasoning Layer standardizes detected events using a common event schema and uses a Neo4j knowledge graph containing a synthetic data center topology, deterministic correlation rules, and a Bayesian network to estimate the likelihood that multiple events belong to the same incident. The Response Layer retrieves relevant information from 13 approved public documents and uses grounded specialist agents to generate evidence-based explanations and recommendations supported by citations. The Decision Layer applies transparent, versioned, deterministic severity rules to prioritize incidents, ensuring that severity classification is not determined by a language model, and that operational actions remain under human control. Finally, the Presentation Layer provides an interactive web dashboard featuring a Monitoring Center, Scenario Lab, analytics, an audited human review queue, and an optional AI Investigator to support incident investigation and decision-making.

The project is particularly relevant because modern data centers face increasingly complex and interconnected security, safety, and operational risks. Detecting individual events independently may not provide sufficient situational awareness or reveal the relationships between seemingly unrelated alerts. DC-GUARDIAN addresses this challenge by correlating evidence across monitoring domains, identifying meaningful incident patterns, reducing isolated or redundant alerts, and prioritizing incidents based on explainable reasoning. Through its integration of knowledge graphs, Bayesian reasoning, and Multi-Agent RAG, the system provides security and maintenance personnel with unified incident triage, contextual explanations, maintenance insights, and actionable recommendations. This

approach is intended to improve operational awareness, support earlier risk identification, and enable faster, more informed decision-making while maintaining human oversight.

To evaluate the effectiveness of the proposed system, DC-GUARDIAN was tested using 24 held-out labelled scenarios. The evaluation demonstrated that cross-domain correlation reduced the number of alerts requiring operator review from 65 to 44, representing a reduction of approximately one-third. Furthermore, 14 out of 18 incidents were presented completely within a single alert, with no isolated alerts remaining for those incidents. The system also improved alert prioritization, with every HIGH-severity alert corresponding to a real incident in the evaluation scenarios. The average precision of the alert queue increased from 0.73 to 0.81, with a 95% confidence interval for the improvement ranging from +0.02 to +0.19. The overall conclusion remained consistent when ChatGPT's labels of the same blind pack were used as an alternative ground truth. These results indicate the potential of explainable cross-domain reasoning to reduce operator workload and improve incident prioritization compared with isolated single-domain monitoring.

Despite these promising findings, the project has several limitations. The evaluation scenarios and data center topology are synthetic, the primary incident labels were developed by the project team, and the system has not yet been deployed or validated using real operational facility data. Therefore, DC-GUARDIAN should be considered a functional industrial prototype rather than a production-ready monitoring solution. Nevertheless, the system demonstrates how explainable reasoning, evidence-grounded AI assistance, transparent severity classification, and human-in-the-loop decision-making can be integrated into a unified monitoring platform.

The main goals of DC-GUARDIAN are to develop a unified platform that monitors physical security, personnel safety, environmental conditions, cybersecurity, and equipment health; accurately identify authorized and unauthorized personnel and detect PPE compliance; identify suspicious SSH activities and abnormal environmental conditions; assess hard-drive failure risks to support proactive maintenance; use a Neo4j knowledge graph, deterministic correlation rules, Bayesian reasoning, and Multi-Agent RAG to connect events across monitoring domains; and provide unified incident triage, transparent prioritization, evidence-based explanations, and actionable recommendations through an interactive dashboard. Additionally, the project aims to evaluate whether cross-domain correlation improves alert quality and reduces operator workload compared with conventional isolated monitoring approaches. Ultimately, DC-GUARDIAN seeks to demonstrate a proactive, explainable, and human-centered approach to intelligent data center protection and maintenance by transforming fragmented monitoring information into meaningful, prioritized incidents.

2. Project Background / Market Context

Data centers have become a fundamental part of modern digital infrastructure, supporting cloud computing, enterprise applications, artificial intelligence, financial services, telecommunications, and other mission-critical services. As organizations become increasingly dependent on data center availability and reliability, protecting these facilities requires more than traditional physical access control or cybersecurity tools. Modern data centers must simultaneously address physical security, personnel safety, environmental hazards, cybersecurity threats, equipment health, operational reliability, and maintenance requirements. The growing scale and complexity of data center infrastructure have also increased the need for continuous monitoring, intelligent threat detection, and integrated security solutions. Current market research indicates strong growth in the data center physical security market, driven by the expansion of hyperscale facilities, increasing cyber-physical threats, and the adoption of AI-based monitoring and detection technologies.

Protecting modern data centers involves monitoring several interconnected risk domains. Cybersecurity monitoring focuses on detecting suspicious activities such as repeated SSH login failures, unauthorized remote access, and potential server break-in attempts. Physical security monitoring identifies

individuals entering restricted areas and determines whether they are authorized to access specific facilities or server halls. Personnel safety monitoring ensures compliance with the required Personal Protective Equipment (PPE) standards in equipment areas. Equipment health monitoring identifies potential hardware failures, including hard-drive failure risks, to support preventive maintenance. Environmental monitoring tracks conditions such as temperature, humidity, and cooling performance that may affect equipment reliability and operational continuity. Although these domains are closely related in practice, they are often monitored through independent tools and technologies.

A major challenge addressed by DC-GUARDIAN is the fragmentation of existing monitoring and security systems. Data centers commonly rely on Security Information and Event Management (SIEM) platforms for cybersecurity logs, video surveillance and access-control systems for physical security, Data Center Infrastructure Management (DCIM) platforms for environmental and cooling conditions, and asset-management systems for equipment maintenance. Although these systems can effectively detect events within their respective domains, they may operate independently and provide limited understanding of the relationships between events occurring across different systems. Therefore, the fundamental problem is not necessarily a lack of detection capability, but rather a lack of connection and contextual understanding between the detected events.

For example, a facial recognition system may identify an unauthorized individual entering Server Hall A, while a safety monitoring camera detects that the individual is not wearing the required PPE. A few minutes later, a cybersecurity monitoring system may detect repeated unauthorized SSH login attempts or a server break-in attempt within the same hall. When these events are viewed independently, they may appear as separate low- or medium-priority alerts on different monitoring consoles. However, correlating them based on location, time, and relevant event relationships could reveal a more serious physical and cybersecurity incident requiring immediate attention. Similarly, an environmental monitoring system may detect abnormal temperature and humidity, while an equipment monitoring system identifies an increased risk of hard-drive failure. Without cross-domain correlation, these alerts may remain disconnected even when environmental conditions could be contributing to equipment degradation or operational failure.

This fragmentation creates several operational challenges. First, it contributes to alert overload, where a single underlying incident generates multiple notifications across different monitoring platforms, increasing the workload of security and maintenance personnel. Second, it results in missed context and reduced situational awareness, as operators may fail to recognize that events occurring within the same location or timeframe are related. Third, it can lead to slow and inconsistent incident triage, because prioritization may depend on which monitoring console an operator is observing and the individual's experience or judgment rather than a unified assessment of the available evidence. Furthermore, disconnected environmental and equipment monitoring information can make proactive maintenance more difficult, potentially delaying the identification of developing hardware problems and their contributing conditions.

The consequences of ineffective monitoring, delayed incident response, and operational failures can be financially significant. According to Uptime Institute's Annual Outage Analysis 2025, 54% of operators reported that their most recent significant outage cost more than US$100,000 [1]. The analysis also highlights the importance of human and organizational factors in data center reliability, reporting that human error is involved in approximately two-thirds to three-quarters of outages and that 80% of operators believe their most recent incident could have been prevented through improved management and processes [2]. These findings emphasize the importance of improving operational procedures, incident visibility, and decision support. They also demonstrate why effective monitoring should extend beyond individual event detection to include contextual analysis, explainable prioritization, and coordinated human response.

The current market is increasingly moving toward AI-enabled, integrated, and centralized security platforms. Research on data center physical security highlights the growing use of Artificial Intelligence (AI) and Machine Learning (ML) for intelligent video analytics, predictive threat detection, automated anomaly recognition, and real-time monitoring. There is also increasing convergence between physical security and Internet of Things (IoT)-enabled environmental monitoring, allowing multiple security layers to be managed through centralized platforms [3]. These developments reflect a broader industry movement toward intelligent monitoring systems that can identify emerging risks, improve operational efficiency, and support more proactive security management.

Another important market trend is the convergence of cybersecurity and physical security. Recent industry research identifies the integration of biometric authentication, AI-powered video analytics, IoT sensors, environmental monitoring, and cybersecurity technologies as an emerging direction for data center protection. Predictive maintenance and continuous equipment monitoring are also becoming increasingly important as operators seek to prevent infrastructure failures, reduce downtime, and maintain service availability [4]. However, integrating multiple detection technologies into a centralized platform does not automatically provide meaningful incident correlation. To achieve more effective situational awareness, monitoring systems must also establish relationships between events, interpret their combined significance, and communicate their findings in a transparent and understandable manner.

From a research perspective, technologies such as Knowledge Graphs, Retrieval-Augmented Generation (RAG), multi-agent AI systems, computer vision, anomaly detection, and probabilistic reasoning provide opportunities to move beyond conventional alert-based monitoring. Knowledge graphs can represent relationships between people, devices, sensors, servers, physical locations, and incidents, enabling systems to identify contextual connections between events. RAG can retrieve relevant operational procedures, maintenance documentation, incident-response guidance, and safety regulations to support evidence-grounded explanations and recommendations. Multi-agent architecture allows specialized AI agents to analyze information from different monitoring domains and contribute to a unified incident assessment. In addition, Bayesian reasoning can estimate the probability that multiple events belong to the same underlying incident, while deterministic correlation and severity rules can support consistent and explainable incident prioritization.

DC-GUARDIAN addresses these challenges by introducing an intelligent reasoning layer that connects evidence across physical security, personnel safety, cybersecurity, environmental monitoring, and equipment health. Rather than presenting every detected event as an isolated alert, the system standardizes incoming evidence, identifies relationships using a Neo4j Knowledge Graph and deterministic correlation rules, and shows a Bayesian probability for each link as context. Multi-Agent RAG then supports the interpretation of correlated incidents by retrieving relevant operational knowledge and generating evidence-based explanations with citations. Incident severity is determined through transparent, deterministic rules rather than language-model judgment, ensuring that prioritization remains consistent, auditable, and explainable. Importantly, the system maintains a human-in-the-loop approach, where operators retain responsibility for reviewing incidents and making operational decisions, and no autonomous response actions are executed.

Therefore, DC-GUARDIAN fits within the broader market transition from isolated monitoring solutions toward integrated, AI-driven cyber-physical security and maintenance platforms. Its distinctive contribution is not simply bringing different monitoring tools into one dashboard but connecting the evidence they generate through explainable cross-domain reasoning. By combining AI-based detection, sensor monitoring, Neo4j Knowledge Graph technology, Bayesian reasoning, deterministic incident correlation, and Multi-Agent RAG within a unified framework, the project aims to reduce fragmented alerts, improve incident prioritization, support proactive maintenance, and enhance overall situational awareness. Ultimately, DC-GUARDIAN seeks to demonstrate how

intelligent reasoning and human-centered decision support can transform disconnected monitoring events into meaningful, actionable incidents, enabling faster, more consistent, and better-informed responses to complex data center security and operational risks.

2.1 Market landscape

The data-center physical security market alone was valued at US$2.56 billion in 2025 and is forecast to reach US$7.53 billion by 2034, a compound annual growth rate of 12.7% [5]. Industry analysts point to three trends that favor an integrated reasoning layer:

- **Convergence of physical and cyber security,** with access control, video analytics, IoT sensors and cybersecurity managed from central platforms.
- **AI-enabled monitoring,** including video analytics, anomaly detection and predictive maintenance.
- **Pressure to reduce outages,** most of which involve human or process failure rather than a missing sensor [1].

3. Literature Review / Market Research

This section reviews the research behind each part of DC-GUARDIAN and the products already sold to data-center operators. The literature shows that each individual technique is mature; the gap is in connecting them across domains in a way an operator can trust. The market review shows the same pattern: strong tools exist for each domain, but each sees only one slice of the facility.

3.1 Literature review

Alert overload and alert correlation. Security operations teams already struggle with the volume and quality of alerts. In interviews with security operations center (SOC) analysts, Alahmadi et al. found that analysts see very high false-positive rates, that most false alarms are triggered by legitimate activity, and that validating them by hand leads to "alarm burnout and eventually desensitization" [6]. They recommend alarms that are reliable, explainable, analytical, contextual and transferable. Alert correlation is the long-standing research answer to this problem. Valeur et al. describe a complete correlation process for intrusion-detection alerts: alerts are first normalized into a common format, duplicates are fused, related alerts are grouped into threads and sessions, and the results are prioritized by their likely impact [7]. DC-GUARDIAN applies the same idea, but across security, safety and operations domains rather than within cybersecurity alone, and its Common Event Schema plays the role of the normalization step.

Knowledge graphs for context. A knowledge graph represents entities and the relationships between them as a graph that can be queried and reasoned over, so that facts are connected rather than stored as separate records [8]. Sikos surveys knowledge graphs built for cybersecurity and shows that a graph can integrate data from heterogeneous sources, make relationships and attack paths visible, and support reasoning over connected entities [9]. These properties match the questions DC-GUARDIAN must answer, such as whether a person recognized by a camera is authorized for the zone that camera monitors, or whether an SSH anomaly and a hot sensor concern the same server hall. This is why the project models the data center, its people and its evidence in Neo4j.

Predictive maintenance in data centers. Ahmed et al. study predictive maintenance in data centers and show that failures of computing and power equipment have a significant operational and financial impact; they argue for analyzing component failures to act earlier and extend equipment life [10]. Botezatu et al. predict disk replacement from SMART telemetry: using data from more than 30,000 disks over 17 months, they identify the SMART parameters that correlate with replacement and predict it 10 to 15 days in advance [11]. Their work supports two choices in DC-GUARDIAN: using real SMART data (Backblaze Drive Stats) rather than a generic machine dataset and predicting failure a few

days ahead rather than detecting it on the day. Both studies treat maintenance on their own; DC-GUARDIAN connects drive risk to environmental conditions in the same hall.

Anomaly detection without labels. Attacks are rare compared with normal activity, and the SSH logs used in this project carry no attack labels. Isolation Forest detects anomalies by isolating observations with random trees: unusual points need fewer splits to isolate, which makes the method fast and memory-efficient without modelling normal behavior in full [12]. Autoencoders take a complementary approach. They learn to reconstruct normal data, and a high reconstruction error marks an anomaly; Sakurada and Yairi show that this non-linear dimensionality reduction can detect anomalies that linear methods such as principal component analysis miss [13]. DC-GUARDIAN combines both explicit rules for known attack patterns and accepts a high-confidence anomaly only when at least two of the three agree.

Computer vision for safety and access. YOLO frames object detection as a single regression problem: one network predicts bounding boxes and class probabilities for the whole image in one pass, which makes real-time detection practical [14]. The Ultralytics YOLOv8 family used in this project follows this single-stage design [15]. For faces, RetinaFace detects and aligns faces in a single stage [16], and ArcFace trains embeddings with an additive angular margin loss so that images of the same person lie close together and different people lie far apart [17]. Because recognition becomes a distance comparison between embeddings, a new person can be enrolled without retraining, and a face that is far from every enrolled person can be rejected as unknown.

Reasoning under uncertainty. Bayesian networks represent a set of variables and their conditional dependencies and update the probability of a hidden cause as evidence is observed [18]. They suit DC-GUARDIAN's correlation problem, where several imperfect clues (time gap, location, domain mechanism and detector strength) must be combined into one degree of belief that two events are related, and where each clue's contribution can be shown to the operator.

Grounded language models. Retrieval-augmented generation (RAG) combines a language model with passages retrieved from an external document collection, so answers can be drawn on, and cite, specific sources [19]. Grounding matters because language models are known to hallucinate Ji et al. survey how generated text can be fluent yet unsupported by, or contradictory to, its source, and review methods to measure and reduce this [20]. DC-GUARDIAN therefore uses RAG only to explain evidence, restricts it to approved documents, checks every citation against what was retrieved, and adds a post-generation judge for unsupported claims.

Research gap. The reviewed work offers mature techniques for each part of the problem: alert correlation within cybersecurity [7], knowledge graphs for security context [9], drive-failure prediction [11], anomaly detection [12], [13], visual detection [14]–[17], probabilistic reasoning [18] and grounded generation [19]. It does not combine physical security, safety, cybersecurity, maintenance and environmental evidence for one facility, nor does it address the concern raised by SOC analysts that alarms should be explainable and contextual [6]. DC-GUARDIAN does not propose a new detection algorithm. Its contribution is to integrate these techniques into one governed reasoning pipeline, in which rules and graph context link the evidence, language models only explain it, and a person decides, and to evaluate whether this improves on isolated single-domain monitoring.

3.2 Market research: existing solutions

Data-center operators already buy mature products for each monitoring domain. Table 1 compares the main categories with DC-GUARDIAN.

Table 1. Existing solutions and how DC-GUARDIAN differs

Product	Primary focus	Where DC-GUARDIAN differs
Schneider Electric EcoStruxure IT (DCIM) [21]	Real-time visibility of power, cooling and IT infrastructure, alarm management and environmental monitoring	Confirms the demand for centralized, proactive monitoring. DC-GUARDIAN adds a knowledge graph, rule-based cross-domain correlation and grounded explanations that link facility alarms with security evidence.
Vertiv Trellis [22]	Centralized monitoring of critical infrastructure, environmental conditions, power systems and facility alarms	Offers comparable infrastructure monitoring; DC-GUARDIAN focuses on reasoning across telemetry, physical access and cybersecurity evidence.
IBM Maximo Asset Performance Management [23]	Asset health monitoring, predictive maintenance and work management supported by AI	Shows the demand for condition-based maintenance. DC-GUARDIAN treats drive risk as one input to cross-domain incidents rather than a separate maintenance workflow.
Splunk Enterprise Security [24] and Microsoft Sentinel [25]	Security information and event management (SIEM): log collection, threat detection and investigation	Correlate events within cybersecurity. DC-GUARDIAN makes the correlation of cybersecurity events with physical location, environmental conditions and equipment health its primary focus.
Genetec Security Center [26]	Unified physical security: access control, video surveillance, communications and event management	Strong on physical security; DC-GUARDIAN adds cybersecurity, equipment and environmental evidence, knowledge-graph relationships and knowledge-grounded explanation.

None of these products combines all five domains with explainable, rule-governed correlation. DC-GUARDIAN is designed to consume their outputs through adapters rather than replace them: a site keeps its SIEM, video and DCIM tools, and DC-GUARDIAN adds the reasoning layer that links their alerts into explained incidents.

3.3 Business case

The value proposition is fewer, better-explained alerts for the operations team, with an audit trail of every human decision. DC-GUARDIAN would be adopted site by site, starting with a short pilot. The Solution tiers in Table 2 are indicative and would be priced with pilot customers, based on the savings measured during the pilot.

Table 2. Indicative adoption tiers

Tier	Scope
Pilot	One site, up to 3 zones, connectors for existing tools, 3 months, ending with a review of the alerts saved
Site	One site, up to 10 zones, all five domains, operator seats and the review workflow
Enterprise	Multiple sites, custom connectors, on-premises deployment, support SLA

Illustrative saving. Using assumed figures, a site raising 200 cross-domain alerts a day, at 5 minutes of triage each, would review about 64 fewer alerts a day at the 32% reduction measured in Section 8. That is about 5.3 analyst-hours a day, or roughly 1,950 hours a year, before counting any outage avoided through earlier detection. The alert volume and triage time are assumptions for illustration, not measurements from a real site.

Barriers to commercialization. The PPE training data (SH17) is licensed for non-commercial use only, the YOLOv8 detector is released under AGPL-3.0, and the face-model weights are for research use. A commercial release would need commercially licensed data and models, consented staff enrolment and a biometric-data policy.

4. Project Scope

The project sets out to answer one research question stated in the proposal: Does combining multi-domain evidence with graph context and explicit rules improve incident correlation and prioritization compared with isolated single-domain monitoring?

DC-Guardian is an intelligent, multi-domain monitoring and decision-support framework for data center environments. It integrates five evidence-generation domains: face recognition, personal protective equipment (PPE) detection, predictive maintenance, SSH anomaly detection, and environmental monitoring.

The framework applies six complementary intelligent reasoning techniques to transform heterogeneous monitoring outputs into actionable insights:

Evidence-based inference: domain-specific models and monitoring components recognize authorized personnel, assess PPE compliance, identify equipment degradation, flag anomalous SSH activity, and detect environmental irregularities.

Knowledge graph reasoning: a Neo4j knowledge graph represents relationships among devices, services, locations, and infrastructure components.

Topology-aware reasoning: physical and logical dependencies are used to correlate events deterministically and identify potentially affected assets.

Retrieval-augmented reasoning (RAG): incident explanations are grounded in approved operational knowledge and retrieved system context.

Agent-based reasoning: grounded AI agents interpret correlated evidence, summarize incidents, and support investigative workflows.

Rule-based reasoning: deterministic decision rules translate detected conditions and correlated evidence into consistent response recommendations that follow predefined operational policies rather than relying solely on generative model outputs.

Together, these techniques form a hybrid reasoning framework that combines data-driven inference, graph-based contextual reasoning, retrieval-grounded language-model reasoning, and deterministic decision logic. A centralized dashboard supports human reasoning by presenting alerts, evidence, relationships, explanations, and recommended actions through a unified interface.

The project focuses on validating the feasibility of integrating heterogeneous monitoring outputs into a unified reasoning and decision-support architecture within a research environment. It provides topology-aware contextual reasoning and evidence correlation; it does not claim causal inference or reliable root-cause determination. The current controlled evaluation maps these outputs to a synthetic topology and injects scenario timing explicitly. This supports reproducible reasoning experiments while keeping model performance claims separate from cross-domain correlation claims.

4.1 Objectives and how each was measured

Table 3. Objectives, success measures and outcomes

Objective	Success measurement	Outcome
Detect evidence in five domains with frozen, independently evaluated models	Each detector evaluated once on a locked test set	Met: all five evaluated (Section 8.1)
Representing the data center and its people as a knowledge graph	Contract tests on schema, topology mapping and ingestion	Met: 24 of 24 Reasoning contracts pass
Correlate evidence across domains, with explanations	Precision, recall and F1 against labelled scenarios	Met: crisp rule F1 0.78, Bayesian F1 0.86 on the test split
Improve on isolated monitoring	Alerts shown, incidents shown whole, queue average precision	Met: 32% fewer alerts, average precision +0.08 (Section 8.2)
Ground explanations in approved knowledge	Retrieval recall and precision; citation and boundary checks	Met: Recall@3 96.1%; unsupported claims flagged by a judge
Keep humans in control	No autonomous action; audited verdicts and reviews	Met: enforced by design and tests
Respond fast enough for operations	Per-stage latency	Met: uploads processed in 0.6 to 3.1 s with persistent workers
Be used by operators	Task success and System Usability Scale	3 participants completed 5 of 7 core tasks without help; SUS 60.8 (OK); severity rules not yet understood (Section 8.7)

4.2 Intelligent Reasoning

Evidence layer (five frozen components)

- Face Recognition: ArcFace embeddings with RetinaFace detection, used as controlled identity evidence for physical security.
- PPE Detection: YOLOv8m at 1,024 px (PPE-v2m, the runtime model; PPE-v1 YOLOv8n stays selectable) detecting person-level PPE against a project compliance policy requiring a helmet and a safety vest.
- Predictive Maintenance: Temporal Random Forest v2, assessing seven-day hard-drive failure risk from SMART telemetry.
- SSH Anomaly Detection: SSH Detector v1, combining rules, Isolation Forest, an Autoencoder, and explicit OpenSSH security signals.
- Environmental Monitoring: a deterministic (non-ML) component converting configured sensor and hardware-telemetry conditions into structured assessments.

Reasoning layer

- A common event schema with validation.
- A controlled synthetic DC-01 topology and mappings.
- Neo4j knowledge-graph ingestion.
- Deterministic, idempotent correlation of any event pair and of connected groups of three or more events across all five domains, with persistent correlation objects.

Response layer

- Operational RAG over approved knowledge, with abstention when knowledge is insufficient.
- Grounded specialist agents, routing, and cross-domain synthesis.

Decision layer

- Deterministic severity and decision rules (DCG-DECISION-v1 and v2), covering single events in every domain and unified groups.

Presentation layer

- A FastAPI backend (role-based access control, Evidence workflows, correlation, human-review audit) and a React/Vite frontend.
- Monitoring Center, Scenario Lab, Analytics Dashboard, and Human Review Queue.
- Authenticated human-review audit records with local audit-integrity verification.
- An opt-in, read-only AI Investigator (research prototype).
- A research-only Bayesian estimate shown beside each correlation pair.

Verification and security

- A single-command full check (check_all.py), a 24-contract Reasoning suite, runtime smoke tests, and CI with security scans.
- A scoped pre-dashboard security assessment (dependency audit, secret scanning, static analysis, RAG source-fetch controls, model artifact integrity, Cypher query boundaries, prompt-injection controls).

5. Data Collection and Preparation

DC-Guardian adopts a hybrid multimodal data strategy because no single public benchmark provides synchronized physical-security, safety, cybersecurity, environmental, and predictive-maintenance observations from one facility. The implemented model-development sources include Backblaze Drive Stats for storage-drive maintenance, Loghub OpenSSH for SSH behavioral modeling, a controlled face-identification dataset organized into enrollment, validation-known, validation-unknown, test-known,

and test-unknown splits, and SH17 for PPE object detection. The SH17 project split contains 8,099 images: 6,479 training images, 810 development-validation images, and a locked 810-image final test. Synthetic topology, authorization records, asset placement, and scenario timing provide the relational context required by Neo4j.

5.1 Data and knowledge sources

Table 4. Data and knowledge sources

Domain	Source	Use	Real or synthetic
Maintenance	Backblaze Drive Stats [27], 2025 Q2 to 2026 Q1 (17.1 million drive-days in the 2026 Q1 holdout)	Train on Q2 and Q3, select on Q4, test once on Q1	Real
Cybersecurity	Loghub OpenSSH [28]	Parse, features and model development	Real
Cybersecurity	SSH honeypot dataset, 145,425 events [29]	External behavioral validation only	Real
Safety	SH17 PPE dataset [30], 8,099 images (6,479 train, 810 validation, 810 locked test)	Train and test the PPE detector	Real
Safety	Safety Helmet and Reflective Jacket [31] (1,571 test images); CHV [32] (1,330 images)	External PPE tests; the Kaggle training split was also used for PPE-v3	Real
Physical security	Face Verification Dataset [33]: 46 images of 13 people	5 enrolled as staff; known and unknown faces for validation and the locked test	Publicly sourced face photographs used for controlled academic evaluation; not consented employee data.
Environmental	Controlled timestamped temperature and humidity readings from dedicated environmental sensors and server hardware telemetry, processed using deterministic thresholds.	Deterministic threshold assessment	Controlled test inputs
Knowledge	13 approved public documents from NIST, CISA, OSHA and U.S. DOE [34]–[41]; the project's own response policy [42] is planned	RAG knowledge base	Real (approved sources)

Domain	Source	Use	Real or synthetic
Infrastructure	DC-01 topology: 3 zones, 2 racks, 4 servers, 3 cameras, 3 sensors, 1 chiller, people and authorizations	Knowledge graph	Synthetic
Evaluation	40 labelled cross-domain scenarios (102 events, 85 candidate pairs)	Correlation, Bayesian and severity evaluation	Synthetic

SH17 was the primary dataset for PPE model development and evaluation. Additional public datasets were used to assess generalization and to develop the PPE-v3 experimental model. PPE-v2m remains the selected runtime model, while PPE-v3 was evaluated as an alternative and was not deployed.

5.2 Preparation

- **Strict evaluation boundaries.** Machine-learning detectors were developed using defined training, validation, and locked-test partitions appropriate to their respective datasets. Model configurations and operating thresholds were selected during development, with final evaluations performed on held-out data. The maintenance dataset uses a chronological split to prevent future observations from entering training. Environmental monitoring does not require a training split because it applies predefined deterministic thresholds, which are verified through contract tests. The PPE-v2m runtime selection is explicitly documented as test-informed rather than an independent validation-based choice.
- **Feature engineering.** SSH logs are parsed and grouped by source IP into 5-minute windows (failed logins, invalid users, root attempts, break-in warnings and more). Drive histories become 33 temporal features from 8 SMART signals. PPE images are evaluated at the detector's input size, and person-to-PPE association is done per detected person.
- **Normalization.** Every detector output becomes a Common Event Schema event with its original assessment and provenance kept unchanged.
- **Synthetic placement.** Events are mapped onto DC-01 assets and given scenario times. They are labelled as synthetic placement, never presented as observations from one real site.

5.3 Labelled scenarios

The detectors are evaluated on their own datasets in Section 7, but the reasoning layer needs something those datasets do not contain: abnormal events from different domains placed in the same data center, with a known answer about which events belong together. No public dataset provides this, so the project built its own labelled scenarios.

What a scenario is. A scenario is a 120-minute window on the synthetic DC-01 topology containing 2 to 4 abnormal events from different domains. Some scenarios represent simulated incidents in which events are labelled as belonging to the same underlying incident, such as a cooling failure that heats a server hall and stresses its drives, an intrusion followed by a console attack, a person without PPE tailgating through a door, or an engineer dispatched to a failing drive. Others are coincidences, where events happen close together but are unrelated. Some are deliberately hard cases designed to tempt a correlator into a false link. The 40 scenarios contain 102 events in total.

What is labelled. Every pair of events in a scenario that the correlation rules could consider is marked related or not related. There are 85 of these candidate pairs. Incidents are the groups formed by related pairs. These labels are the ground truth for the correlation results in Section 8.

Splits. 16 scenarios form the development split, used to design and tune the rules. The other 24 form the held-out test split, kept aside and scored once for the reported results.

Two versions. Scenario set v1 was written and labelled by the project operator, the team member who authored the scenarios. Set v2 replaced eight scenarios whose alerts (water leak, smoke, airflow) the deployed environmental detector cannot produce, using temperature and humidity readings instead. Because the scenarios changed, no v1 label was carried over; all 40 were relabeled from scratch.

Blind labelling. A scenario's title and narrative reveal its intended answer, so the relabeling used a blind pack. The pack holds the 40 scenarios under neutral IDs in shuffled order, showing only the events: no titles, stories, split or model output. It also includes a short facility brief, a glossary and a form listing the 85 candidate pairs. The scenario author labelled the blind pack without access to scenario titles, narratives, split assignments, or model predictions during the labelling process. These labels serve as the reference ground truth for the reported evaluation. Although the blind procedure reduces direct narrative influence, the labels are not independently established because the same operator authored the scenarios.

Sensitivity check. ChatGPT labelled the same blind pack. It agrees with the operator on 73% of the 85 pairs, with a Cohen's kappa of 0.46. Kappa corrects agreement for chance: 0 means chance level and 1 means perfect agreement, so 0.46 is moderate. Section 8 repeats the main results with ChatGPT's labels as ground truth.

6. System Design

DC-GUARDIAN is an intelligent monitoring and reasoning system for one controlled, synthetic data center (DC-01). It takes evidence from five independent monitoring domains (physical security, safety, cybersecurity, maintenance and environment) and turns it into a short list of explained, ranked incidents for a human operator.

The design is a five-layer pipeline: Evidence, Reasoning, Response, Decision and Presentation. Each layer adds context to what it receives, and no layer rewrites what came before. Detectors produce evidence, symbolic reasoning links it, language models only explain it, deterministic rules rank it, and a person makes every final decision. This separation is the central design idea: each kind of reasoning is used only for the job it does best, and every handover between layers is a tested contract.

This section answers the three parts of the brief in order. Section 6.1 sets out the academic and market requirements the design had to meet. Sections 6.2 to 6.4 present the architecture, with diagrams, then each layer and the domain model in detail. Section 6.5 explains how the reasoning techniques were compared and chosen, Section 6.6 follows one incident through every layer, and Sections 6.7 and 6.8 give the design principles and what changed since the proposal.

6.1 Design requirements

The design had to satisfy two sets of requirements at once. The academic requirements come from the Intelligent Reasoning Systems certificate: the system must apply and combine reasoning techniques from its courses (Machine Reasoning, Reasoning Systems and Cognitive Systems), explain its conclusions, and be evaluated rigorously. The market requirements come from business background and market research in Sections 2 and 3: data-center operators already own strong single-domain tools, suffer from too many disconnected alerts, and work in a regulated, safety-critical setting where every decision must be traceable to a person.

Table 5. Design requirements and how the design meets them

ID	Requirement	Where it comes from	How the design meets it	Where it is checked
A1	Combine several reasoning techniques	IRS brief (MR, RS, CGS)	Knowledge graph and rules (machine reasoning), Bayesian network (reasoning under uncertainty), machine-learning perception and retrieval-grounded language models (cognitive systems)	Contract tests for every technique (Section 7)
A2	Explain every conclusion	IRS brief; research goal	Each link carries its reason and what it does not prove; rules name the rule that fired; specialists cite retrieved passages	Dashboard; citation and claim checks
A3	Evaluate against a baseline, on unseen data	Research question	Correlation compared with isolated single-domain alerts on 24 held-out scenarios; each detector scored once on a locked test	Section 8
A4	Keep claims precise	Research integrity	Synthetic data labelled as synthetic; correlation never claims cause, identity or compromise	Wording rules and post-generation judge
M1	Fewer, more meaningful alerts	Alert fatigue; outage costs (Section 2)	Related alerts are grouped into one explained incident	Alerts 65 to 44 (Section 8)
M2	Work with the tools sites already have	Market research: each tool sees one slice (Section 3)	Every detector feeds one Common Event Schema through an adapter, so a new source needs only a new adapter	Adapter contract tests
M3	Predictable, auditable severity	Compliance and operator trust	Versioned deterministic rules; no language model sets severity	Rule tests; saved Decisions stay reproducible
M4	Human control, no autonomous action	Safety-critical facility	Every Decision is HUMAN_REVIEW with autonomous action disabled	Decision contract tests
M5	Protect people and data	Biometrics; security tooling	Local storage of face embeddings; opt-in AI Investigator with per-request operator consent; role- and zone-based authorization with deny-by-default access; outbound data minimization and pseudonymization; and hash-chained audit records. These controls support the	Access matrix over all 99 API operations

ID	Requirement	Where it comes from	How the design meets it	Where it is checked
			research prototype but do not constitute production security certification.	
M6	Fast enough for an operator	Day-to-day use	Detector models stay loaded with persistent workers	SSH analysis 7.3 s to 0.56 s

6.2 Architecture overview

The architecture is shown at three levels of detail:

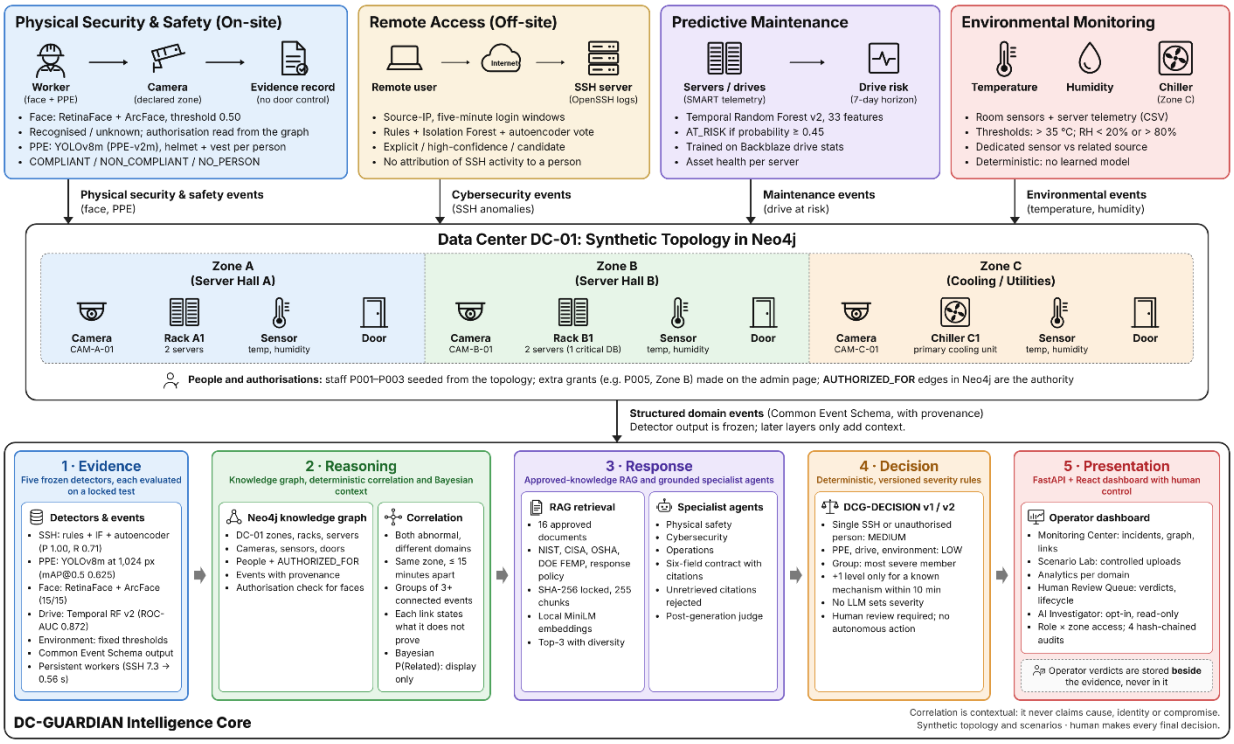


Figure 1- System overview: five evidence sources, the DC-01 knowledge graph and the intelligence core

Figure 1 presents the overall DC-GUARDIAN system, illustrating the five monitoring domains, the synthetic DC-01 data-center topology, and the integrated intelligence core.

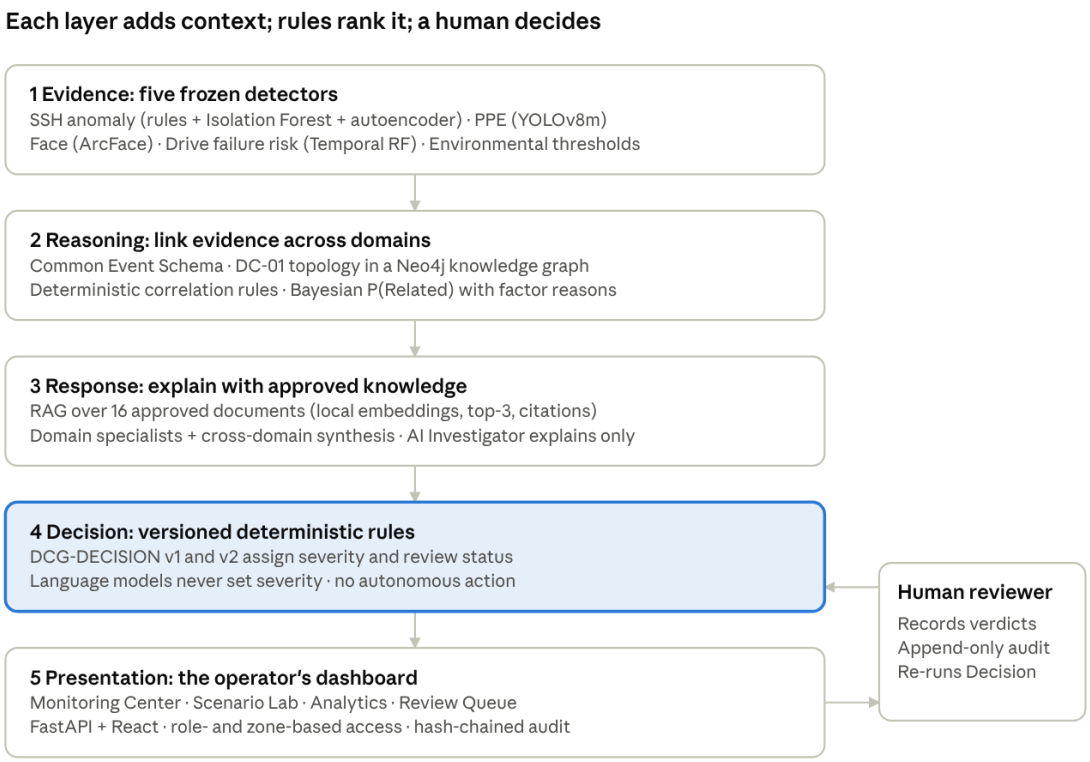


Figure 3- DC-GUARDIAN architecture: five layers and the human review loop

Figure 2 illustrates the five-layer architecture, showing how evidence passes through Reasoning, Response, Decision, and Presentation, with human review as the only mechanism for revising decision outcomes.

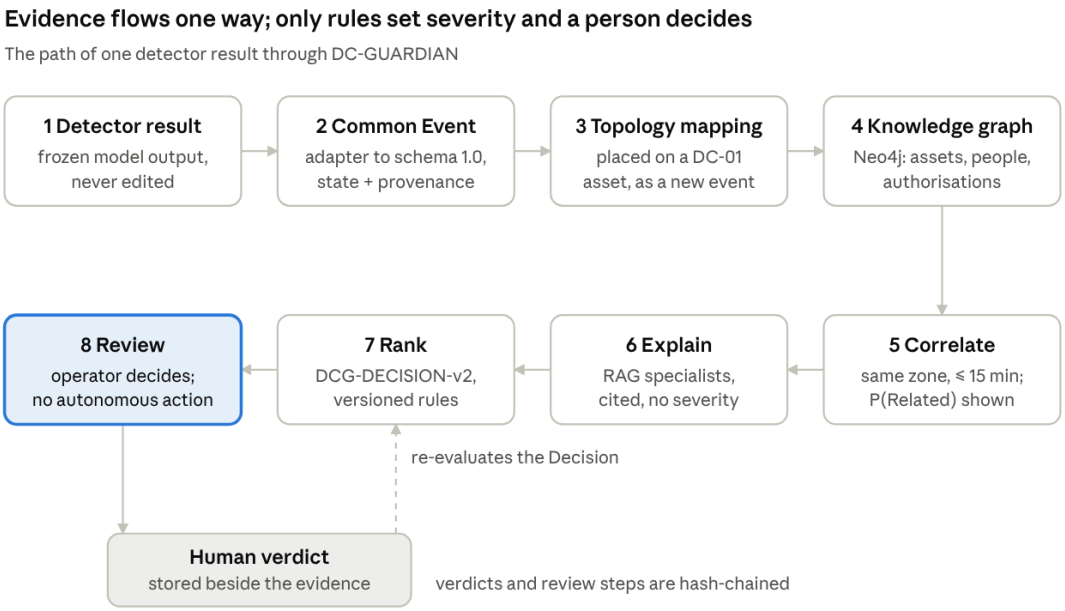


Figure 5 - Detailed DC-GUARDIAN evidence-processing workflow from detector output to human review.

Figure 3 presents the detailed evidence-processing workflow, from the original detector output through Common Event Schema normalization, topology mapping, knowledge graph integration, correlation, grounded explanation, deterministic severity assignment, and human review.

Three properties of this flow matter for the rest of the design. First, it is one-way: no later step can change a detector result, so the original evidence is always available for audit. Second, severity is set in exactly one place, step 7, by versioned rules; the correlation step adds context, and the explanation step adds words, but neither can raise or lower a level. Third, the only backward path is human: an operator's verdict is stored next to the evidence, and the Decision is recomputed from it by the same rules, so the reason for every change is recorded.

6.3 Layer-by-layer design

Table 6 summarizes each layer's job, what it receives and produces, and what it is deliberately not allowed to do. The paragraphs after it explain the reasoning behind each layer.

Table 6. Each layer's purpose, inputs, outputs and limits

Layer	Purpose	Input	Output	Deliberately does not
1- Evidence	Detect abnormal conditions in each domain	SSH logs, camera images, SMART telemetry, sensor CSV	One Common Event per detector result	Correlate, rank or retrain at runtime
2- Reasoning	Place evidence in context and link related events	Common Events, DC-01 topology, authorizations	Graph nodes, correlation links and groups, P(Related)	Claim cause, identity, presence or compromise
3- Response	Explain evidence in plain language with sources	A group or event and its graph context; approved documents	Six-field specialist assessment with citations	Set severity or create correlations
4- Decision	Assign severity and the review requirement	Events, groups, linkage signals, human verdicts	A versioned Decision (severity, rules fired, HUMAN_REVIEW)	Use a language model; act autonomously
5- Presentation	Let people see, question and decide	Everything above	Dashboard views, verdicts, review lifecycle, audit records	Edit evidence; hide the rule or source behind a result

Layer 1: Evidence

Five detectors, one per domain, turn raw data into evidence: an SSH anomaly detector (rules, an Isolation Forest and an autoencoder that vote), a PPE detector (YOLOv8m), face recognition (RetinaFace with ArcFace embeddings), a drive failure-risk model (Temporal Random Forest v2) and an environmental threshold monitor. Each evidence component was developed and validated according to its underlying method. The machine-learning detectors use frozen model configurations and documented evaluation procedures, while environmental monitoring applies fixed temperature and humidity thresholds. The PPE-v2m runtime model was selected following additional locked-test

analysis, and this test-informed selection is explicitly recorded as an evaluation limitation. Detector assessments and provenance are preserved when converted into the Common Event Schema. Freezing matters for the architecture because every later layer can then trust that the same input always produces the same evidence, and because a detector can be improved later as a new version without silently changing past results. An adapter per detector converts its native output into the Common Event Schema (Section 6.4) without changing its conclusion. The detectors run in persistent worker processes, so models load once, which cut SSH analysis from 7.3 s to 0.56 s and face analysis from 17.8 s to 3.1 s.

Layer 2: Reasoning

This layer answers the question a flat alert list cannot: which of these alerts belong together, and what do we know about them? It has three parts.

- **Topology mapping and knowledge graph**. Each event is placed on a DC-01 asset (a server, a camera's zone, a sensor) and stored in Neo4j with links to the infrastructure and people around it. Mapping creates a new scenario event rather than editing the original, and camera evidence is mapped only to its zone, because a camera cannot place a person at a specific server. The graph also provides authorization context: a recognized person is checked against the AUTHORIZED_FOR links of the zone where the camera saw them.
- **Deterministic correlation**. Two events are linked only if both are abnormal, they come from different domains, they share a declared zone and they are at most 15 minutes apart. Linked events that form a connected set of three or more become a group (an incident candidate). Every link records its reason and a statement of what it does not prove.
- **Bayesian network**. For each linked pair, a small Bayesian network estimates P(Related) from four observed factors: where the events are relative to each other (same zone or not), the time gap, whether the domain pair has a known physical mechanism, and how strong each detection is. Its probabilities were set by expert judgement and frozen before evaluation. It is shown to the operator as context only; the rules still decide which events are linked.

Layer 3: Response

This layer explains the evidence; it never changes it. A retrieval-augmented generation (RAG) pipeline searches 13 approved public documents (NIST, CISA, OSHA and U.S. DOE FEMP guidance), which are hash-locked, split into 255 passages and embedded locally. Three specialist agents (physical safety, cybersecurity and operations) each receive only their domain's evidence and retrieved passages and must answer in a fixed six-field contract: assessment, supported findings, recommended considerations, grounding status, citations and limitations. Citations to passages that were not retrieved are rejected, a synthesis step may not upgrade a claim (for example from "recognized" to "identified as the attacker"), and a post-generation judge flags overclaims. The opt-in AI Investigator provides read-only explanations of existing evidence and correlation records. Each request requires explicit operator consent and is subject to authorization, outbound data minimization, source-reference validation, targeted field-level consistency checks, and additional safeguards against unsupported claims. These mechanisms improve the traceability and reliability of generated explanations but do not guarantee complete factual correctness. The Investigator cannot modify detector evidence, establish new correlations, assign severity, or initiate autonomous actions.

Layer 4: Decision

Severity comes only from versioned, deterministic rules. DCG-DECISION-v1 covers the original saved SSH, operational and pipeline Decisions and is kept unchanged so those remain reproducible. DCG-DECISION-v2 covers single events in every domain and correlated groups: an SSH anomaly or an unknown or unauthorized person is MEDIUM; PPE non-compliance, a drive at risk and an abnormal environment are LOW. A group starts at its most severe member and is raised by one level only when its members show a known mechanism (environment with maintenance, cybersecurity with physical security, or physical security with safety) within 10 minutes. Co-occurrence alone never escalates.

Every Decision records the rules that fired, has response mode HUMAN_REVIEW, and sets autonomous action to false. Because the v2 group rule was chosen on the development split and validated on the operator's blind labels, it is always shown with that caveat.

Layer 5: Presentation

A FastAPI backend and a React dashboard give the operator four workspaces: the Monitoring Center (incidents, evidence, graph view and links), the Scenario Lab (controlled uploads for every detector), Analytics (outcomes per domain) and the Human Review Queue (verdicts and the investigation lifecycle). Human verdicts (confirm, override, inconclusive) are stored beside the detector's result and trigger a deterministic re-evaluation of the Decision. Access is controlled by role and zone and denied by default, and four hash chains (group reviews, evidence verdicts, Investigator calls and the investigation lifecycle) make the audit trail tamper evident.

6.4 Domain model and Common Event Schema

The data center is modelled as a Neo4j knowledge graph. The core infrastructure node types include DataCenter, Zone, Rack, Server, Camera, Sensor, Equipment, AccessPoint, Person, and SourceIP. Event nodes represent normalized detector observations, while correlation records connect related evidence. Monitored hardware components are referenced through asset identifiers within the Common Event Schema and maintenance evidence. Relationships such as LOCATED_IN, HOSTED_BY, MONITORS, OBSERVED_BY, TARGETS, IDENTIFIES, AUTHORIZED_FOR and CONTRIBUTES_TO support contextual reasoning across infrastructure, people, and events.

Table 7. Node types in the DC-01 knowledge graph

Node type	What it represents in DC-01
DataCenter	The site, DC-01
Zone	Server Hall A, Server Hall B, and Cooling and Utilities (Zone C)
Rack, Server	Racks A1 and B1, each holding two servers; SRV-B1-01 is the critical database server
Asset / asset reference	A monitored hardware component, such as a drive associated with a server.
Camera, Sensor, AccessPoint	One camera, one environmental sensor and one door per zone
Equipment	The primary chiller in Zone C
Person	Staff with their zone authorizations
SourceIP	An address seen in SSH logs
Event	One normalized detector result, placed on DC-01
Correlation	A saved link or group between events

Table 8. Relationships in the DC-01 knowledge graph

Relationship	Meaning
Zone PART_OF DataCenter; Rack LOCATED_IN Zone; Server LOCATED_IN Rack	Where things are
Asset HOSTED_BY Server	Which server a drive belongs to
Camera or Sensor MONITORS Zone, door or equipment	What each device can see or measure
AccessPoint CONTROLS_ACCESS_TO Zone	Which door leads to which zone
Person AUTHORIZED_FOR Zone	Who may enter where
Event TARGETS Server or Asset; OBSERVED_BY Camera; IDENTIFIES Person; ORIGINATED_FROM SourceIP; LOCATED_IN Zone	What an event is about
Event CONTRIBUTES_TO Correlation; Correlation CONCERNS infrastructure	Which events form a link or group, and what it is about

The topology is a versioned JSON contract marked SYNTHETIC; it seeds the graph and the initial authorizations. Authorizations added later on the dashboard's administration page live only in Neo4j, which is the authority for who may enter where.

The initial DC-01 topology contains three zones, two racks, four servers, three cameras, three environmental sensors, three access points, one cooling unit, and three predefined staff identities (P001–P003) with zone-specific authorizations. Additional recognized identities may be enrolled in the face-recognition system, but recognition alone does not establish authorization. Zone access is determined from the authorization relationships available in the knowledge graph.

Example: the authorization check. When face recognition recognizes a person, the system does not ask the model whether that person is allowed in. It follows the graph instead: the face event is OBSERVED_BY a camera, the camera MONITORS a zone, and the system checks whether the recognized person has an AUTHORIZED_FOR link to that zone. If the link is missing, the event's authorization context is UNAUTHORIZED; if the face is not recognized at all, it is UNKNOWN_PERSON. The result is explainable as a path, can be checked by anyone with access to the graph, and changes the moment an administrator grants or removes access, with no retraining.

The Common Event Schema

Five detectors with five different output formats could not be reasoned about together, so every result is converted into one Common Event Schema (version 1.0) by a per-detector adapter. The schema is validated at every event.

Table 9. Common Event Schema fields

Field	Content	Why it is needed
event_id, schema_version	Unique ID and schema version	Traceability and safe evolution of the format
timestamp, window	When it happens, or the window it covers	Time-based correlation

Field	Content	Why it is needed
domain	CYBERSECURITY, PHYSICAL_SECURITY, SAFETY, MAINTENANCE or ENVIRONMENTAL	Cross-domain rules and severity tables
event_type, source	What kind of result, and which component and version produced it	Knowing which frozen model spoke
entities	Person, source IP, asset, server, camera	Linking to graph nodes
location	Data center, zone, rack, access point	Placing the event on DC-01
assessment	The detector's own state (for example EXPLICIT_SECURITY_EVENT, AT_RISK, NON_COMPLIANT) and confidence	The original conclusion, kept unchanged
evidence	Domain-specific details (votes, scores, model version, PPE items)	Explanation and review
provenance	Source type, whether the placement is synthetic, original timestamp, scenario ID	Honesty about where the data came from

The assessment field preserves the detector's classification, confidence, and applicable scores, while the evidence field retains domain-specific observations such as SSH anomaly features, PPE detections, face-matching results, or maintenance indicators. Provenance records the original source and whether synthetic topology or scenario mapping was applied, allowing downstream reasoning to use contextualized evidence without altering the originating detector result.

Two rules keep the schema trustworthy. Adapters copy the detector's conclusion as it is, so a later layer cannot accidentally strengthen it. And provenance always records whether an event's placement on DC-01 is synthetic, so no result is ever presented as an observation from a real site.

6.5 Choosing the reasoning techniques

No single technique could meet all the requirements in Section 6.1. Machine-learning models are good at perception but cannot explain a severity level; language models explain fluently but can invent facts; rules are transparent but brittle where evidence is uncertain. The design therefore asks, for each job the system must do, which technique fits that job best, and gives each technique only that job.

Selection criteria

Every option was judged against the same six criteria, taken from the requirements:

1. **Explainability:** can an operator or examiner see why it produced its output? (A2, M3)
2. **Reproducibility and auditability:** does the same input always give the same output, and can it be versioned and tested? (A3, M3)
3. **Data it needs:** does it need labelled training data that this project does not have?
4. **Handling uncertainty:** can it express degrees of belief rather than only yes or no?
5. **Risk of overclaiming or acting:** could it state something the evidence does not support, or take an action on its own? (A4, M4)
6. **Fit to the domain:** does it match the shape of the problem (images, logs, time series, relationships, text)?

The thinking process, decision by decision

1. Detecting abnormal conditions in each domain. Each domain has different data, so each detector was chosen separately rather than forcing one model on all five.

- *SSH logs.* The Loghub OpenSSH log has no attack labels, which ruled out a supervised classifier as the main detector. Unsupervised models (Isolation Forest and an autoencoder) can learn what normal login windows look like without labels, and explicit rules catch well-known attack signatures. Because each method misses different things, they vote: two or more votes give a high-confidence event. The consensus reached precision 1.000 and recall 0.713 on 280 locked scenarios.
- *PPE images.* Checking helmets and vests is an object-detection problem, so a YOLO detector was a natural fit. The proposal's YOLOv8n was replaced by YOLOv8m at 1,024 px after three experiment rounds (mAP@0.5 rose from 0.545 to 0.625); v2m was kept over v3m because it misses fewer real violations, which matters more for safety than a slightly higher average score.
- *Faces.* With only a few photos of five enrolled people, training a classifier was not realistic, and a classifier cannot say, "I don't know this person". Pretrained ArcFace embeddings with a cosine-distance threshold handle both: a face is matched to the nearest enrolled person or rejected as unknown. The 0.50 threshold sits in the gap between known (≤ 0.33) and unknown (≥ 0.79) faces on validation, and the locked test scored 15 of 15.
- *Drives.* Failure is predicted from how SMART readings change over time. A Temporal Random Forest on 33 engineered features was compared with a GRU challenger; the GRU was rejected because it did not beat the forest, and the forest is easier to explain through its features. It reached ROC-AUC 0.872 in a later, unseen quarter.
- *Environment.* Temperature and humidity limits are already defined by operating guidance, and there is no labelled anomaly data to learn from, so fixed thresholds are the honest choice. No accuracy is claimed for them.

2. Representing context. Relational tables could store the same facts, but every question ("is this person authorized for the zone where this camera saw them?") would need several joins and the answer could not be shown as a path. A language model's memory was rejected because it cannot be audited. A knowledge graph stores relationships as first-class data, answers these questions by following links, and lets the dashboard draw the path that justifies a conclusion.

3. Deciding which events belong together. This was the central reasoning choice, and five options were compared (Table 10).

Table 10. Options compared for deciding which events belong together

Option	Explainable	Reproducible	Needs labelled data	Expresses uncertainty	Risk of overclaiming
Language-model judgement	Low: reasons can be invented	Low	No	Informally	High
Learned classifier	Low to medium	Medium	Yes, many labelled pairs	Yes	Medium
Fuzzy rules	Medium	High	No	Yes, as membership degrees	Low

Option	Explainable	Reproducible	Needs labelled data	Expresses uncertainty	Risk of overclaiming
Deterministic rules	High	High	No	No	Low
Bayesian network	High: one factor per reason	High once frozen	No (expert probabilities)	Yes, as a probability	Low

The language model was ruled out because a correlation it proposed could not be reproduced or audited, and a learned classifier because there were no labelled event pairs at design time (the 40 labelled scenarios were built later, for evaluation). That left rules, fuzzy logic and a Bayesian network. Deterministic rules were made the **primary** method: they need no training data, behave the same on every run, and each link can state exactly which conditions it met. Their weakness is that they are all-or-nothing, so a second method was added to express how strong a link is. A Bayesian network was preferred to fuzzy rules because the evidence comes from independent factors that should combine multiplicatively, which a Bayesian network does by design and explains as one likelihood ratio per factor (for example, "a known mechanism makes this link 2.2 times more likely"). Fuzzy rules would need membership functions and a combination operator that are harder to justify for this kind of evidence.

The Bayesian network assumes that the observed factors are conditionally independent given whether events are related. This simplifying assumption supports transparent likelihood-ratio explanations but may not capture dependencies between time, topology, and operational mechanisms in real data-center incidents.

The experiments then decided the Bayesian network's role. On the held-out scenarios it scored a higher F1 than the rules (0.857 against 0.784), but the 95% interval of the difference (−0.040 to +0.190) includes zero, so the gain is not significant. It is therefore shown to the operator as context and does not decide which events are linked.

4. Explaining evidence. Fixed text templates are safe but cannot adapt to new combinations of evidence, and a free language model adapts but can invent procedures or facts. Retrieval-augmented generation sits between them: the model writes in plain language but may only use passages retrieved from approved documents, every citation is checked against what was actually retrieved, and a judge flags overclaims after generation (it caught 4 of 5 overclaims, with 1 false alarm in 35 correct answers). Splitting the work across three domain specialists keeps each prompt small and each answer within one field of expertise.

5. Assigning severity. A language model was ruled out because the same evidence could get a different level on another day, and a learned model because the scenarios carry no severity labels to learn from. A weighted score was considered but its weights would be arbitrary and hard to defend to an auditor. Versioned deterministic rules were chosen: each level can be read, tested and traced to the rule that set it. For groups, four candidate escalation rules were declared before testing and compared on the development split; the selected rule (known mechanism within 10 minutes) was then checked once on the test split, where groups it escalated were genuine incidents far more often than groups it did not.

6. Taking action. Automatic responses (locking a door, blocking an IP address) were ruled out for a prototype evaluated on synthetic data in a safety-critical setting. The system recommends and records; a person decides.

Summary: one technique per job

Table 11. One technique per job

Job	Technique chosen	Family of reasoning
Detect abnormal conditions	Isolation Forest, autoencoder and rules; YOLOv8m; ArcFace; Temporal Random Forest; thresholds	Machine learning and perception; rules
Represent context	Neo4j knowledge graph	Knowledge representation
Link events	Deterministic correlation rules	Rule-based reasoning
Express how likely a link is	Bayesian network	Probabilistic reasoning under uncertainty
Explain	RAG with grounded specialist agents	Language understanding and agents
Rank	Versioned decision rules	Rule-based reasoning
Decide	Human review	Human in the loop

The detailed mapping of these techniques to the certificate's courses is given in the appendix.

6.6 Worked example: the Zone A incident

The following controlled synthetic scenario illustrates how DC-GUARDIAN processes and reasons about multiple abnormal events. It demonstrates the intended end-to-end workflow rather than documenting an incident observed in a real data center.

Table 12. The Zone A incident, layer by layer

Layer	What happens	Result
1- Evidence	Face recognition matches P005 (distance below 0.50). The PPE model finds one compliant and one non-compliant person. The SSH detector's votes give a high-confidence anomaly. Each adapter writes a Common Event.	Three events, each with its own detector state and provenance
2- Reasoning: graph	Each event is placed on DC-01: the face and PPE events are OBSERVED_BY CAM-A-01, which MONITORS Zone A; the SSH event TARGETS SRV-A1-02 in Rack A1, Zone A. P005 has no AUTHORIZED_FOR link to Zone A.	Face event marked UNAUTHORIZED from the graph
2- Reasoning: correlation	Every pair is abnormal, from different domains, in Zone A and within 15 minutes, so the three links form one connected group. Each link states what it does not show, for example that the person seen is not shown to be the SSH actor.	One group of three events, with P(Related) per pair

Layer	What happens	Result
3- Response	Specialists explain the group with retrieved access-control, PPE and incident-response guidance, each claim citing a retrieved passage.	Grounded explanation; no severity
4- Decision	The most severe member is MEDIUM. Physical security with cybersecurity is a known mechanism, within 10 minutes, so rule L2 raises the group one level.	HIGH, HUMAN_REVIEW, autonomous action disabled
5- Presentation	The group tops the Monitoring Center. The operator reads the links and explanations, records a verdict with a rationale, and resolves the investigation.	Verdict stored beside the evidence; chained audit record

Without correlation, the operator would have seen three separate alerts on three consoles (one LOW and two MEDIUM), each easy to dismiss on its own. With it, they see one HIGH incident with its reasons, while every original detector result stays unchanged and every claim stays within what the evidence shows.

6.7 Design principles

Five principles run through every layer. Each one exists to protect a requirement from Section 6.1, and each is enforced by code and tests rather than by convention.

- **Evidence is immutable**. Detector output and model files are never edited. Specialist explanations, Investigator answers and human verdicts are stored beside the evidence, never in place of it, so the original result is always available for audit, and a detector can be replaced by a new version without rewriting history. (Protects A4, M3.)
- **Correlation is context**, **not proof**. Sharing a zone and a time window never establishes identity, cause, physical presence or compromise, and every link says what it does not show. A person seen near a server is not presented as the person who logged into it. (Protects A4.)
- **Language models explain; rules decide**. No model output can set severity, create a correlation or change a detector result. Language models are used only where their strength (plain-language explanation) is needed, and their weakness (invention) can be checked. (Protects M3, A2.)
- **Humans decide.** Every Decision has response mode HUMAN_REVIEW and autonomous action set to false. The system ranks and explains; it never acts on the facility. (Protects M4.)
- **Everything is versioned and audited**. Schemas, topology, detectors, knowledge bases and rules carry versions, rules change only as a new version so saved Decisions stay reproducible, and reviews, verdicts, Investigator calls and investigation steps are hash chained. (Protects M3, M5.)

6.8 Changes from the proposal

The original proposal was written before full implementation was completed. It described a system with three monitoring domains, three core reasoning techniques (a knowledge graph, RAG and specialized agents), a correlation agent that would decide which events form an incident, rules and optimization for decisions, and a Streamlit dashboard. Building and testing the system changed several of these choices. Table 13 compares the proposal with the final system and gives the reason for each change.

Table 13. Changes from the original proposal

Area	In the original proposal	In the final system	Why it changed
Project focus and title	"Data Center Physical & Cybersecurity Correlated with Predictive Maintenance"	Multi-domain reasoning for data-center security, safety and operations	The work grew to five domains and a full reasoning pipeline, so the title was broadened
Domains	Three: physical security, cybersecurity, equipment monitoring	Five: physical security, safety, cybersecurity, maintenance, environmental	Separating PPE from access and drives from room conditions lets severity tables and mechanisms treat each correctly
Core reasoning techniques	Knowledge graph, RAG, specialized agents	Knowledge graph, deterministic correlation rules, Bayesian network, RAG with specialist agents, versioned decision rules	Agents alone could not give reproducible, auditable links or severity (Section 6.5)
Who decides that events are related	A correlation agent	Deterministic rules; the Bayesian network adds P(Related) as context	An agent's judgement could not be reproduced or tested; rules can
Role of the agents	Assess incidents and recommend a response	Explain evidence only, in a fixed six-field contract with checked citations; they cannot set severity or create links	Keeps language models away from decisions while keeping their value for explanation
Severity	A field in every event, plus escalation rules	Removed from events; set only by DCG-DECISION-v1/v2 in the Decision layer	Severity in the event would have let detectors or adapters decide it; one place for severity makes it auditable
Optimization	Technician assignment and maintenance scheduling	Not built; future work	Effort went into validating the reasoning core; optimization would need real scheduling data
Maintenance data and model	AI4I 2020 as a proxy, with Isolation Forest as a baseline	Backblaze drive statistics; Temporal Random Forest v2 (33 features) with a GRU challenger	AI4I is synthetic factory-machine data; Backblaze gives real hard-drive failures, which match a data center
Face recognition	OpenCV-based detection and recognition	RetinaFace detection with ArcFace	Pretrained embeddings are far more accurate and can reject unknown faces

Area	In the original proposal	In the final system	Why it changed
		embeddings and a 0.50 distance threshold	
PPE detection	Fine-tuned YOLOv8 for hard hats, vests, gloves and more	YOLOv8m at 1,024 px (PPE-v2m) with a helmet-and-vest policy	Experiments showed small items needed a larger model and image size; the policy covers the two items required in this scenario
SSH detection	Train and compare anomaly-detection approaches	Rules, Isolation Forest and an autoencoder that vote	No attack labels in Loghub, so unsupervised models plus rules, combined by voting
Environmental inputs	Humidity, smoke, temperature, vibration and power	Temperature and humidity, with fixed thresholds	These are what the deployed sensor inputs provide; scenario set v2 replaced scenarios that relied on other states
Data flow	A Python event producer replaying data in real time	Controlled uploads through the Scenario Lab, processed by persistent detector workers	Controlled uploads are reproducible and easier to evaluate; persistent workers keep response times low
Dashboard	Streamlit, with monitoring, tickets, topology view, recommendations and chat	FastAPI and React: Monitoring Center, Scenario Lab, Analytics, Human Review Queue and an opt-in AI Investigator	Role-by-zone access, a review workflow, audit and browser testing needed a full web stack; the chat became a consent-based, read-only Investigator
Evaluation	Acknowledged an "evaluation ceiling" from designed scenarios	40 labelled scenarios split into development and held-out test sets, a baseline comparison, bootstrap intervals and locked detector tests	Turned the acknowledged limitation into a measured, honest evaluation
Governance	Not covered	Human verdicts beside evidence, review lifecycle, role-by-zone access, consent, hash-chained audit	Needed for a system whose output people must be able to trust and audit

The core idea of the proposal held throughout: evidence from several independent domains is placed in one knowledge graph of the data center, related events are linked, explanations are grounded in approved documents, and operators see one connected picture instead of separate alerts. What changed is how each part is trusted. Decisions that the proposal left to agents are now made by deterministic, tested rules, language models are limited to explanation, every planned layer was either built and evaluated or explicitly deferred, and the system gained the controls monitoring tool needs before anyone would rely on it.

7. Implementation

All five layers are implemented, integrated and tested in one repository, built with Python 3.12, Neo4j, FastAPI, React and TypeScript, with continuous integration on every change.

7.1 Evidence layer

Table 14. The five evidence detectors

Detector	Method	Output
SSH anomaly	OpenSSH parser; source-IP, 5-minute windows; three votes (rules, Isolation Forest [12], autoencoder) plus explicit OpenSSH break-in signals	EXPLICIT_SECURITY_EVENT, HIGH_CONFIDENCE_ANOMALY (2 or more votes), ANOMALY_CANDIDATE (1 vote), NO_ANOMALY_EVIDENCE
PPE	YOLOv8m [15] at 1,024 px (PPE-v2m at runtime; PPE-v1 YOLOv8n selectable); person-to-PPE association; policy requires helmet and safety vest	COMPLIANT, NON_COMPLIANT, NO_PERSON, with annotated detections
Face	RetinaFace detection [16], ArcFace 512-dimension embeddings [17] via DeepFace [43], cosine match against a private enrolment set at threshold 0.50	RECOGNIZED or UNKNOWN; authorization is decided later from the graph
Drive failure risk	Temporal Random Forest v2: 300 trees, 33 features from 8 SMART signals, threshold 0.45, 7-day horizon	NORMAL or AT_RISK, with failure probability
Environmental	Deterministic thresholds (above 35 °C; humidity below 20% or above 80%), separating dedicated sensors from hardware telemetry	NORMAL or an abnormal state such as HIGH_TEMPERATURE

Model files are checked against frozen SHA-256 digests before loading, and each detector runs as a persistent worker, in its own isolated environment when that environment is installed (otherwise in the main environment, on CPU).

7.1.1 SSH anomaly detector (cybersecurity)

Data. The Loghub OpenSSH log [28]: 655,147 raw lines, all parsed (0 unparsed), of which 542,381 carry a source IP. Loghub has no attack labels, so it was used for parser and feature development and for training the unsupervised models, not to report accuracy. Two controlled labelled scenario sets were then generated: Set A (330 scenarios: 120 normals, 210 attack) for development, and Set B (280 scenarios: 120 normal, 160 attack, 14 attack families with 20 repetitions each), locked before the final evaluation.

Parsing. `parser.py` extracts from each OpenSSH line the timestamp, host, process ID, event type, username, source IP and port, invalid-user flag, success flag, repeat count and original message. Recognized events include failed and successful logins, invalid users, PAM authentication failures, possible break-in warnings, connections, disconnects and missing identification strings.

Behavioral windows. Events are grouped by **source IP and 5-minute window**, giving 8,256 windows. Each window becomes one feature row:

Table 15. SSH behavioral window features

Feature	Meaning	Used by
failed_login_count	Failed authentication attempts	Rules, Isolation Forest, autoencoder
invalid_user_count	Attempts with usernames that do not exist	Rules, Isolation Forest, autoencoder
unique_users	Distinct usernames tried	Rules, Isolation Forest, autoencoder
failure_ratio	Failed attempts divided by all attempts	Isolation Forest, autoencoder
root_attempt_ratio	Share of failed attempts that target root	Rules, Isolation Forest, autoencoder
breakin_warning_count	OpenSSH "POSSIBLE BREAK-IN ATTEMPT" warnings	Explicit signal, Isolation Forest
disconnect_count	Disconnect events	Isolation Forest, autoencoder
no_identification_count	Connections with no SSH identification string	Isolation Forest, autoencoder
successful_login_count, attempt_rate, root_attempt_count, success_after_failures	Context for analysts	Reported, not modelled

Feature selection. Features were chosen for what they say about attacker behavior: brute force (failures, failure ratio), username enumeration (invalid users, unique users), root targeting, and scanning (disconnects, missing identification). The autoencoder leaves out the break-in-warning count because that warning is handled as explicit evidence, and features with zero variance in the normal-reference data were checked and removed.

Models.

- **Rules:** a window is flagged when it shows 10 or more failed logins, 5 or more distinct usernames, or root targeting (at least 5 failures with 80% or more aimed at root).
- **Isolation Forest [12]:** 200 trees, contamination 0.15 (checked with a sensitivity sweep), with a persisted scaler.
- **Autoencoder:** a dense network $7 \to 6 \to 3 \to 6 \to 7$ (ReLU, linear output), trained for 100 epochs with batch size 32 and learning rate 0.001. It learns only normal behavior: the training pool is windows that both the rules and the Isolation Forest call normal, split by source IP so no IP appears in both training and validation. Count features are log-transformed (log1p) and all features are scaled with a RobustScaler. A window is anomalous when its reconstruction error exceeds the 99th percentile of errors on unseen-IP normal validation windows.

Output. The three votes and the explicit warning give an evidence state: EXPLICIT_SECURITY_EVENT (break-in warning), HIGH_CONFIDENCE_ANOMALY (2 or 3 votes), ANOMALY_CANDIDATE (1 vote) or NO_ANOMALY_EVIDENCE. Set A revealed a gap in break-in-warning handling, which was fixed before Set B was locked; Set B results are in Section 8.1.

External check. A honeypot dataset of 145,425 events from 13,897 source IPs [29] confirmed that source-IP windows, root targeting (33.7% of logins), username diversity and connection burst (up to 765 connections from one IP in 5 minutes) matter. It lacks several of the detector's fields, so it is used only as behavioral evidence, not to report accuracy.

7.1.2 Drive failure-risk model (maintenance)

Data. Backblaze Drive Stats [27]: daily SMART readings for every drive in Backblaze's fleet, 2025 Q2 to 2026 Q1, keeping multiple drive models rather than one. The split is strictly chronological: Q2 and Q3 2025 for training, Q4 2025 for validation and all model decisions, and Q1 2026 (17.1 million drive-days, 667 failed drives) as an untouched holdout.

Labelling. Each drive-day is labelled `AT_RISK` if the drive fails 1 to 7 days later and `NORMAL` otherwise. The failure day itself, days after failure, and days whose outcome cannot be known yet (censored) are excluded, so the task is forward-looking rather than same-day detection.

SMART signals. Eight raw attributes were selected because they are present across the retained drive population and relate to known failure modes:

Table 16. SMART attributes used by the drive model

Attribute	Meaning
SMART 5	Reallocated sectors
SMART 198	Offline uncorrectable sectors
SMART 9	Power-on hours
SMART 194	Temperature
SMART 4	Start/stop count
SMART 12	Power-cycle count
SMART 192	Power-off retract count
SMART 193	Load-cycle count

Feature engineering (33 features).

- **Current values:** the 8 SMART readings (8 features).
- **Health indicators:** SMART 5 non-zero, SMART 198 non-zero, SMART 5 increased, SMART 198 increased (4 features).
- **Change over time:** for each signal, the change over 1 and over 7 observations (16 features).
- **Temperature history:** 7-observation mean, minimum, maximum and range, and the current temperature relative to the recent mean (5 features).

Only current and past readings are used, never future ones. Feature groups were added one at a time and kept only when validation PR-AUC improved (Section 8.9, experiment 3).

Class imbalance. Failures are about 0.025% of drive-days. Training keeps every positive row and a deterministic sample of 50 negatives per positive (10,676 positives, 533,800 negatives); this ratio was chosen on validation. Accuracy is not used to select models because a model that always says "normal" would score over 99.9%.

Model. A Random Forest with 300 trees, no depth limit, at least 2 samples per leaf, square-root feature sampling and seed 42, stored together with its preprocessing as one pipeline. A GRU sequence model

(30 days × 8 signals, 14,273 parameters) was trained as a challenger and not adopted, because it did not improve PR-AUC on the same Q4 endpoints (0.016 against 0.023).

Threshold. A probability of 0.45 or more means `AT_RISK`. It was chosen on Q4 validation as a balance between failing drives caught, precision, alert volume and warning lead time, and was not changed after the holdout was seen.

7.1.3 PPE detector (safety)

Data. SH17 [30]: 8,099 workplace photographs annotated with 17 classes (person, head, face, ear, hands, foot, helmet, safety vest, safety suit, medical suit, gloves, glasses, face mask, face guard, earmuffs, shoes and tool). The project split is 6,479 training, 810 validation, and 810 locked test images. Two further public sets were used only for testing: Safety Helmet and Reflective Jacket [31] (1,571 test images after removing 4 near-duplicates of SH17) and CHV [32] (1,330 images, 175 near-duplicates excluded). PPE-v3 also trained on the Kaggle training split, with near-duplicates of any test image removed first.

Preprocessing. SH17 annotations were converted to YOLO format. Because the photos are 2,000 to 6,000 px wide, helmets and vests are only a few pixels across at 640 px, so the v2 candidates were trained on copies resized once to 1,024 px.

Models and training.

- **PPE-v1:** YOLOv8n [15] initialized from COCO-pretrained weights, 640 px, 50 epochs, batch 16, seed 42, on an NVIDIA RTX 3080 Ti.
- **PPE-v2 candidates:** YOLOv8n, s and m at 1,024 px for 60 epochs, seed 42. The selection rule was committed before training: mean mAP@0.5 of the two required classes on validation, with the smaller model winning ties within 0.01, and the winner having to beat PPE-v1. YOLOv8s won on the tie-break.
- **PPE-v2m (runtime):** the YOLOv8m candidate, chosen after the post-hoc analysis and recorded as a test-informed choice.
- **PPE-v3:** YOLOv8m starting from PPE-v2m weights, trained on SH17 plus the Kaggle data, with persons and other classes pseudo-labelled by PPE-v2m because the Kaggle images label only helmets and jackets.

Person-to-PPE association. Each detected PPE item is assigned to the person whose box contains at least 50% of the item's box, choosing the strongest eligible match. Detections are kept at confidence 0.25 or more, with non-maximum suppression at IoU 0.7.

Compliance policy (PPE-POLICY-v1). For the data-center maintenance scenario, each detected person must have a helmet and a safety vest. The image result is `COMPLIANT`, `NON_COMPLIANT` (at least one person missing a required item) or `NO_PERSON`. `NON_COMPLIANT` means the item was not detected; it does not prove the item is absent.

7.1.4 Face recognition (physical security)

Data. The Face Verification Dataset [33] (MIT license for the compilation): 46 images of 13 people, all 100 × 100 px. Five people are enrolled as "staff" (P001 to P005), with 18 enrolment images; the rest are split into validation (7 known, 6 unknown) and a locked test (6 known, 9 unknown). Photos and embeddings are kept out of the repository.

Pipeline. Implemented with DeepFace [43]: RetinaFace [16] detects and aligns the face, and ArcFace [17] produces a 512-dimension embedding. Each probe is compared with the 18 enrolled embeddings by cosine distance; the nearest enrolled identity is accepted if the distance is 0.50 or less, otherwise the result is `UNKNOWN`.

Threshold selection. On validation, known faces had distances between 0.07 and 0.33 and unknown faces between 0.79 and 0.90. The threshold 0.50 was placed inside this gap to keep a margin on both sides; it is a project setting, not a universal ArcFace value.

Authorization. Recognition only says who the person is. Whether they may be in a zone is read from the knowledge graph (`Person -[:AUTHORIZED_FOR]-> Zone`), so a recognized person in the wrong hall produces an `UNAUTHORIZED` result.

7.1.5 Environmental monitoring

Data. Timestamped CSV readings from dedicated room sensors and from server hardware telemetry, uploaded through the Scenario Lab. No public dataset is needed because nothing is learned.

Method. Deterministic thresholds: temperature above 35 °C gives `HIGH_TEMPERATURE`; humidity below 20% gives `LOW_HUMIDITY` and above 80% gives `HIGH_HUMIDITY`. The detector also defines airflow, smoke and water-leak states, but the deployed sensor inputs supply temperature and humidity, which is why scenario set v2 replaced scenarios that relied on the other states.

Source independence. A dedicated sensor is treated as independent evidence; hardware telemetry is marked as a related source because it can overlap with the drive-health evidence from the same server. This stops two readings of the same hardware from counting as two independent signals.

Limitation. The thresholds are initial controlled values and would be aligned with the actual equipment's operating limits in deployment.

7.2 Reasoning layer

The Reasoning layer turns detector outputs into connected, explainable evidence in five steps. Each step has its own contract tests, 24 in total, run in dependency order.

7.2.1 Common Event Schema

One adapter per domain converts detector output into a shared JSON event with these fields: `event_id`, `schema_version`, `timestamp`, optional `window`, `domain`, `event_type`, `source`, `entities`, `location`, `assessment`, `evidence` and `provenance`. The detector's original assessment, scores and model metadata are copied unchanged into `assessment` and `evidence`, and `provenance` records where the event came from, so later layers can add context but cannot silently alter what a model concluded.

7.2.2 Topology mapping

The DC-01 topology is a JSON contract marked `SYNTHETIC`: three zones, two racks with four servers (one critical database server), three cameras, three environmental sensors, three doors, one chiller and three people with zone authorizations. Controlled mapping files place each event on an asset, for example an SSH scenario target on server SRV-B1-01 in Rack B1, Zone B. Mapping creates a new scenario event rather than editing the normalized one, and camera evidence is mapped only to its zone, because a camera cannot place a person at a specific server.

7.2.3 Knowledge graph

Events are written to Neo4j with uniqueness constraints on every node type. Relationships connect evidence to infrastructure, for example an SSH event `TARGETS` a server that is `LOCATED_IN` a rack, which is `LOCATED_IN` a zone; a camera `MONITORS` a zone and a door; a face event `IDENTIFIES` a person who may be `AUTHORIZED_FOR` a zone. Ingestion is idempotent, so replaying an event does not duplicate it.

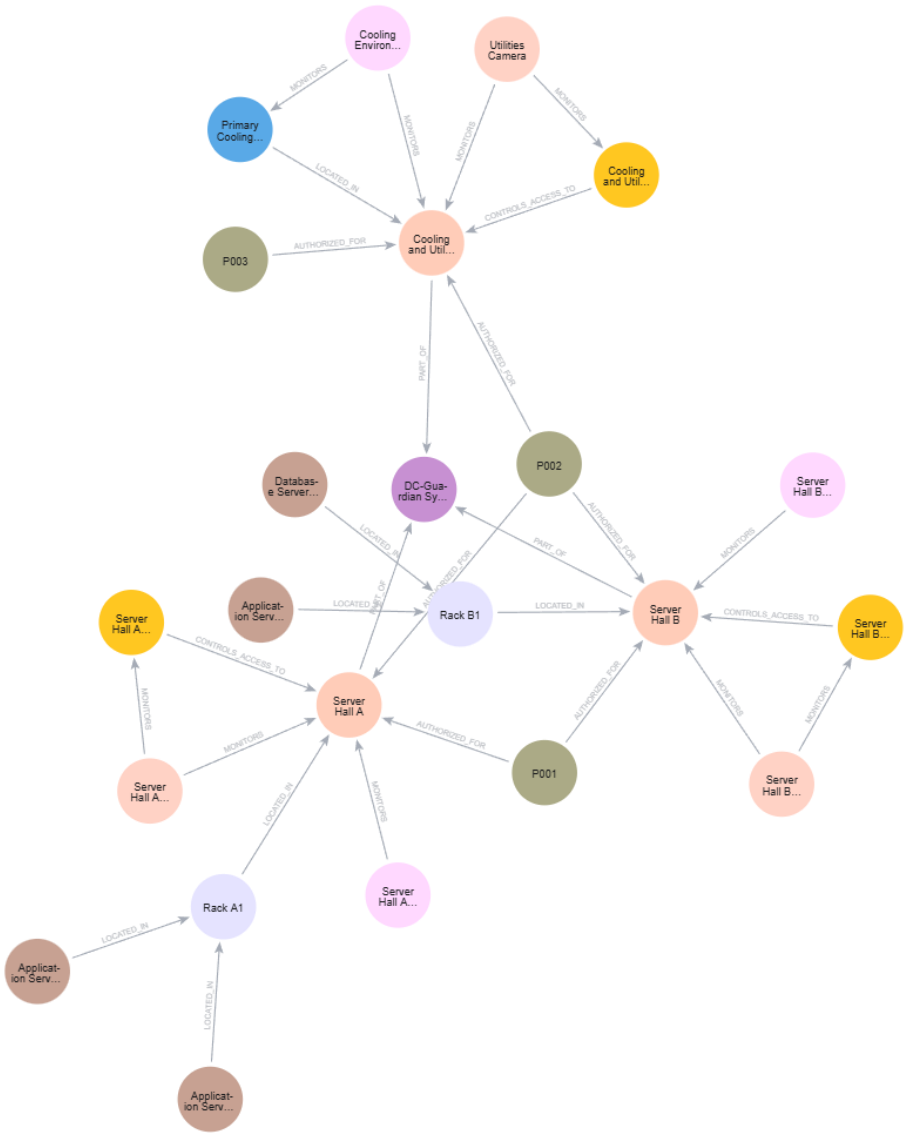


Figure 6 - Overview of Neo4j graph

7.2.4 Deterministic correlation

An event is eligible for correlation only if its state is abnormal: an SSH anomaly state, drive AT_RISK, an abnormal environmental state, PPE non-compliance, an unknown person, or a recognized person who is unauthorized for the zone.

Table 17. Deterministic correlation rules

Rule	Condition
General pair rule	Two abnormal events in different domains, in the same declared zone, at most 15 minutes apart
Maintenance + environment	Independent abnormal assessments in the same zone within 15 minutes

Rule	Condition
PPE + face	Matching image hash, or the same declared camera and capture time
Face + SSH	The same declared zone and time
Scope	The deepest level shared by the events: server, rack or zone
Independence	Dedicated sensors are independent; hardware telemetry is a related source
Groups	Connected sets of three or more correlated events in one zone; no links are invented between members

Every pair and group carries a plain-language statement of what it does **not** establish, for example: "Declared zone/time context only. No identity, source-IP, login, or causal linkage is established."

7.2.5 Persistence

A correlation ID is built from the scenario, scope, shared entity and the sorted event IDs, so saving the same correlation twice merges into one Correlation node. Events link to it with `CONTRIBUTES_TO`.

7.2.6 Bayesian correlation (reasoning under uncertainty)

The crisp rule answers only yes or no. The Bayesian network estimates **P(Related)**, the probability that two abnormal events belong to one incident, from four observed factors that depend on a hidden cause "Related":

Table 18. Bayesian network factors and their states

Factor	States
Time gap	5 minutes or less; 15 or less; 60 or less; more than 60
Topology	Same server; same rack; same zone; linked by the chiller's cooling dependency; unrelated
Domain mechanism	Known (for example heat and drive wear; unauthorized access and a console attack); possible (for example an engineer sent to a failing drive without full PPE); unlikely (for example an SSH anomaly and humidity)
Evidence strength	Both strong; mixed; both weak, using each detector's evaluation (for example SSH consensus is strong, a single vote is weak; drive risk is strong at probability 0.70 or more)

The prior is P(Related) = 0.2. The conditional probabilities are expert estimates recorded with their reasoning in `knowledge_base.json`, which was frozen in its own commit before any evaluation:

Table 19. Expert conditional probabilities (selected states)

Factor state	P (state given related)	P (state given unrelated)
Time gap 5 min or less	0.45	0.04
Time gap is more than 60 min	0.05	0.50
Same server	0.25	0.04

Factor state	P (state given related)	P (state given unrelated)
Same zone	0.40	0.25
Known mechanism	0.65	0.30
Unlikely mechanism	0.07	0.30
Both strong	0.55	0.20
Both weak	0.10	0.40

Inference is exact enumeration in a small dependency-free engine. Each score is explained as likelihood ratios: for example, a room heat alert and a high-risk drive in the same hall nine minutes apart score 0.90 because same zone (×1.6), 15 minutes or less (×3.75), a known mechanism (×2.17) and two strong detections (×2.75) outweigh the low prior. A pair with P(Related) of 0.5 or more is counted as related. The score is shown beside each dashboard pair but does not drive decisions (Section 8.3).

7.3 Response layer

- **Governed retrieval: 13 approved documents**, locked by SHA-256, chunked into 255 passages and embedded locally with all-MiniLM-L6-v2 [44], [45]; top-3 retrieval with domain filtering.
- **Specialists:** physical safety, cybersecurity and operations agents (OpenAI gpt-4.1-mini) must return a fixed six-field assessment whose citations point only to retrieved passages.
- **Safeguards:** a deterministic knowledge-eligibility check returns `INSUFFICIENT_APPROVED_KNOWLEDGE` instead of guessing; boundary claims (identity, compromise, causation, root cause) cannot be upgraded by synthesis; retrieved text is treated as data, tested against prompt injection.

7.3.1 The approved knowledge base

The specialists may only cite a governed corpus of 13 approved public documents from U.S. government authorities. Two further documents (NIST SP 800-34 Rev. 1 and the DOE guide O&M in Federal Facilities) and the project's own response policy are listed in the manifest for future approval but are not in the knowledge base. Each document is recorded in the manifest with its publisher, version, authority type, primary domain, the domains it may serve, approval status and SHA-256 hash.

Table 20. Knowledge-base documents (13 approved; candidates and the planned policy marked)

Document	Type	Primary domain	What it provides
NIST SP 800-61 Rev. 3 [34]	Government guidance	Cybersecurity	Incident response steps and roles
CISA Incident and Vulnerability Response Playbooks [38]	Government playbook	Cybersecurity	Step-by-step response to incidents and vulnerabilities
NIST SP 800-40 Rev. 4 [35]	Government guidance	Cybersecurity	Patch management as preventive maintenance
NIST SP 800-34 Rev. 1 [37]	Government guidance	Shared	Contingency planning and recovery (candidate: listed, not in the knowledge base)

Document	Type	Primary domain	What it provides
NIST SP 800-53 Rev. 5, PE family [36]	Government standard	Physical security	Physical access and environmental protection controls
CISA / ISC Facility Access Control (2020) [39]	Government best practice	Physical security	Facility access control practice
OSHA 29 CFR 1910.132, .135, .137, .147 [40]	Regulation	Safety	PPE general requirements, head protection, electrical protective equipment, lockout/tagout
OSHA 29 CFR 1910.38 [40]	Regulation	Shared	Emergency action plans
U.S. DOE FEMP: O&M Best Practices; O&M Challenges and Solutions [41] (O&M in Federal Facilities is a candidate, not in the knowledge base)	Government guidance and best practice	Maintenance	Operations and maintenance practice for equipment and facilities
U.S. DOE FEMP: Best Practices Guide for Energy-Efficient Data Center Design (2024) [41]	Government guidance	Environmental	Data-center cooling and environmental conditions
DC-GUARDIAN Operational Response Policy v1 [42]	Internal project policy	Shared, all domains	Planned: the project's own response rules, kept separate from external authority; not yet in the knowledge base

How the documents are prepared and retrieved:

1. **Lock:** each approved document is downloaded once, stored as a snapshot and verified against its SHA-256 hash.
2. **Parse and clean:** text is extracted conservatively. OSHA pages are trimmed to their own CFR section, because the official PDFs also print neighboring sections.
3. **Chunk:** documents are split deterministically into passages of about 850 words with 100 words of overlap, giving 255 passages.
4. **Embed locally:** all-MiniLM-L6-v2 creates 384-dimension vectors on the workstation, so no document text leaves the machine during retrieval.
5. **Filter and rank:** only documents approved for the event's domain are searched, and the top 3 passages are selected with controlled diversity so one long document cannot crowd out the others.
6. **Check eligibility:** if no approved document of the required kind exists, retrieval returns `INSUFFICIENT_APPROVED_KNOWLEDGE` rather than letting a similar external source stand in for project policy.

Every retrieved passage keeps its document ID, chunk ID, publisher, page, URL and source hash, and a specialist's citations are rejected if they point to anything that was not retrieved.

7.3.2 Specialist agents and the grounded-assessment contract

Routing is deterministic: the domains already established by the Reasoning layer choose the specialists, in a fixed order, and no language model decides who answers.

Table 21. Specialist agents and their domains

Specialist	Domains	Typical question it answers
Physical safety	Physical security, safety	What do access-control and PPE guidance say about an unauthorized person without a vest?
Cybersecurity	Cybersecurity	What does incident-response guidance recommend for repeated root login failures?
Operations	Maintenance, environmental	What do maintenance and data-center guidance say about heat and drive at risk?

Each specialist receives the structured evidence and the top-3 retrieved passages, calls the model (gpt-4.1-mini, with provider-side storage disabled) and must return exactly six fields:

- `assessment` and `supported_findings`;
- `recommended_considerations` for the human reviewer;
- `grounding_status`: `SUPPORTED`, `PARTIALLY_SUPPORTED` or `INSUFFICIENT`;
- `citations`: chunk and document IDs, which must be among the passages actually retrieved;
- `limitations`.

A deterministic validator rejects missing or extra fields and invented citations. With no retrieved evidence the status must be `INSUFFICIENT`, and an `INSUFFICIENT` answer must state its limitations.

7.3.3 Cross-domain synthesis and boundary claims

When a group spans several specialists, a synthesis step combines their findings. It runs only with at least two specialists covering at least two domains. Four boundary claims are tracked explicitly: identity link, confirmed compromise, causal relationship and root cause. Synthesis may keep a claim an upstream specialist established, but it may never turn a false claim into a true one.

7.3.4 AI Investigator safeguards

The Investigator answers operator questions about one evidence record, a pair, an operational correlation or a group. It is off by default, and every request needs the operator's explicit consent. Safeguards, in the order they apply:

1. **Authorization:** the backend applies the operator's role and zone permissions before selecting any context.
2. **Outbound minimization:** biometric fields are removed, and usernames, IP addresses and person IDs leave only as keyed pseudonyms.
3. **Prompt boundaries:** retrieved text and evidence are passed as data, and the model is instructed not to assign severity, create correlations or take actions.
4. **Source manifest:** each answer lists the evidence records it was given.
5. **Field checks:** numbers and identities quoted in the answer are compared with the source fields.
6. **Claim checks and judge:** sentence-level checks and a post-generation judge flag unsupported claims of presence, causation or PPE absence.
7. **Audit:** Every call is recorded in a hash-chained log; optional conversation history is off by default and kept for 1 to 30 days.

7.4 Decision layer

- **DCG-DECISION-v1:** severity for saved SSH, operational and pipeline Decisions, rising from LOW to HIGH by explicit rules such as cybersecurity evidence or three or more domains.
- **DCG-DECISION-v2:** single-event severities for every domain (a conservative operator-approved policy) and group severity: the most severe member, raised one level only when

members show a known linking mechanism within 10 minutes. Co-occurrence alone never escalates.

- **Re-evaluation:** when a human verdict changes an input, the rules are re-run and the result appended; the original Decision is never modified.

The group rule is short enough to read in full, which is the point of using rules for severity (decision/rules_v2.py):

```
def decide_group(events, rule=None):
    """Severity for 2+ correlated events: most severe member, +1 level on linkage."""
    rule = rule or SELECTED_LINKAGE_RULE          # L2_KNOWN_MECHANISM_WITHIN_10_MIN
    scored = [(e, *event_severity(e)) for e in events]
    concern = [(e, level, rules) for e, level, rules in scored if level]
    base = max((level for _, level, _ in concern), key=SEVERITY_ORDER.get)
    signals = linkage_signals([e for e, _, _ in concern])
    linked = len(concern) >= 2 and LINKAGE_RULES[rule](signals)
    severity = LEVELS[min(len(LEVELS) - 1, SEVERITY_ORDER[base])] if linked else base
```

7.4.1 DCG-DECISION-v1

Used for saved SSH, operational and integrated-pipeline Decisions. Severity starts at LOW and only rises; the highest rule triggered wins.

Table 22. DCG-DECISION-v1 rules

Rule	Condition	Severity
UNAUTHORIZED_PHYSICAL_ACCESS_EVIDENCE	The graph says the recognized person is not authorized for the zone	MEDIUM
CYBERSECURITY_ANOMALY_EVIDENCE	Cybersecurity evidence is present	MEDIUM
CORRELATED_OPERATIONAL_RISK	Both maintenance and environmental evidence are present	MEDIUM
CONFIRMED_COMPROMISE	An upstream boundary claim explicitly confirms compromise	HIGH
MULTI_DOMAIN_CORRELATED_RISK	Three or more domains are present	HIGH

HIGH sets escalation_required. If the specialist's grounding is INSUFFICIENT, the status is EVIDENCE_REVIEW_REQUIRED; otherwise, REVIEW_REQUIRED. Every result carries response_mode: HUMAN_REVIEW and autonomous_action_allowed: false.

7.4.2 DCG-DECISION-v2

Version 2 was added without changing v1, so every saved v1 Decision stays reproducible. It has two parts.

Single events in every domain (a conservative policy approved by the operator, not learned from data):

Table 23. DCG-DECISION-v2 single-event severities

Evidence	Severity
Any SSH anomaly state	MEDIUM

Evidence	Severity
Unknown person in a zone	MEDIUM
Recognized person unauthorized for the zone	MEDIUM
PPE non-compliance	LOW
Drive at risk	LOW
Any abnormal environmental reading	LOW

Correlated groups: the group starts at its most severe member and is raised one level only if its members show a linking mechanism. Co-occurrence alone, however many domains, never escalates. Four candidate linkage rules were declared before comparison on the development split:

Table 24. Candidate group linkage rules

Candidate	Escalate when
L1	The events share a server, or an unauthorized person coincides with cybersecurity evidence
L2 (selected)	The domains form a known mechanism (environment and maintenance; physical security and cybersecurity; physical security and safety) and the events fall within 10 minutes
L3	Any of those mechanisms within 10 minutes
L4	Any of those mechanisms within 5 minutes

A time-only rule was deliberately excluded because it would encode how the scenarios were written rather than a data-center mechanism.

7.4.3 Unified evidence review

For groups, a separate deterministic review lists why a human should look, for example `PPE_EVIDENCE_REQUIRES_SOURCE_REVIEW`, `FACE_SSH_CONTEXTUAL_ASSOCIATION_UNVERIFIED` or `SPECIALIST_GROUNDING_INSUFFICIENT`. It never inherits severity from a standalone Decision.

7.4.4 Human verdicts and re-evaluation

A reviewer can confirm, override or mark any inconclusive evidence. When a verdict changes an input, the same rules are re-run on the human-verified states, and the result is appended as a new record; the original Decision is never edited. If a human clears an input, its severity becomes empty rather than a guessed lower level.

7.5 Presentation layer

- **Monitoring Center:** live status, items needing attention, evidence viewers with received, observed and evaluated times, graph views, correlations and Decisions; At a glance summaries, a Simple view and Full detail for groups, date, zone and severity filters, and a help panel (added after the usability study, Section 8.7).
- **Scenario Lab:** controlled uploads of SSH logs, SMART CSV, sensor CSV, PPE images and face images.
- **Analytics: distributions by domain, severity and zone over a chosen or custom period, review progress, and drill-down to records.**

- **Human Review Queue:** verdicts (confirm, override, inconclusive) on any evidence and a review lifecycle for groups (open, in review, resolved), all hash-chained with external checkpoints; work is sorted into To do, In progress and Completed tabs, with a type filter.
- **AI Investigator:** opt-in, consent per request, read-only; outbound data is minimized and pseudonymized; answers carry source manifests, field checks and a post-generation judge for overclaims.
- **Access control:** deny-by-default roles combined with zone permissions.

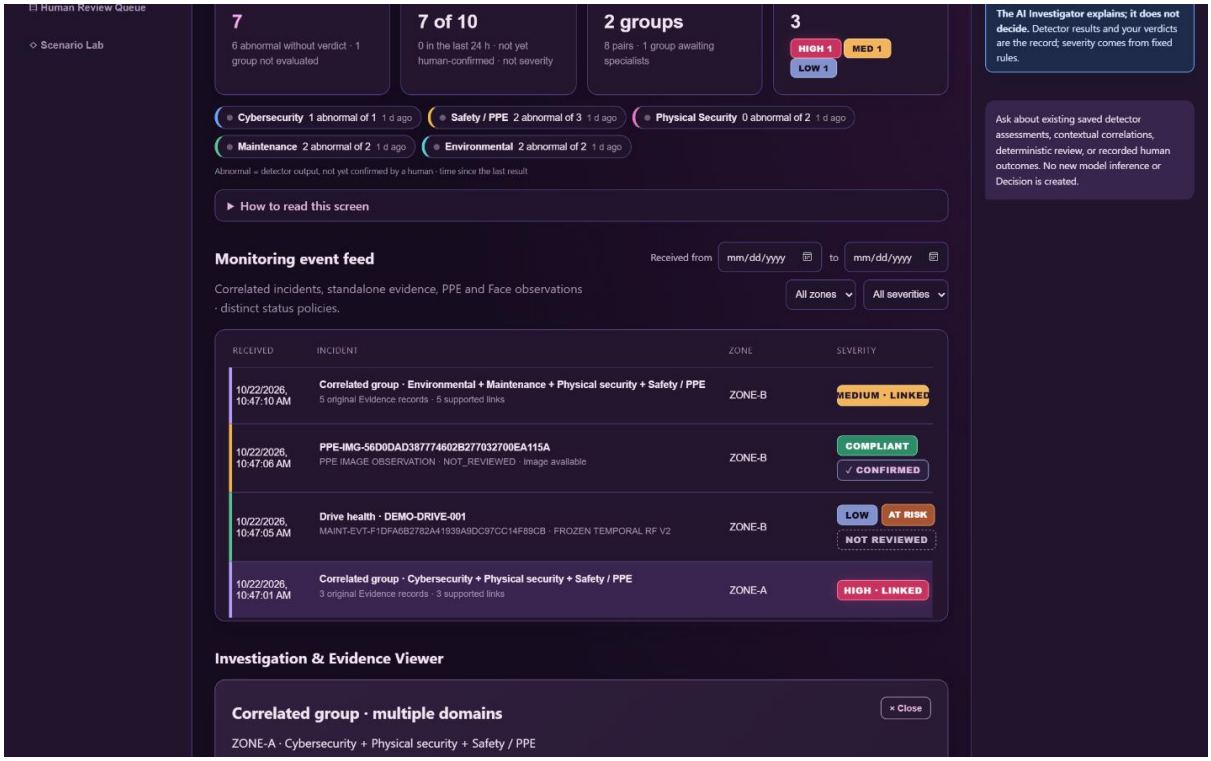


Figure 7. Monitoring Center: summary cards, domain counts ("N abnormal of M"), the help panel and the event feed, with the HIGH · LINKED Zone A group selected

At a glance

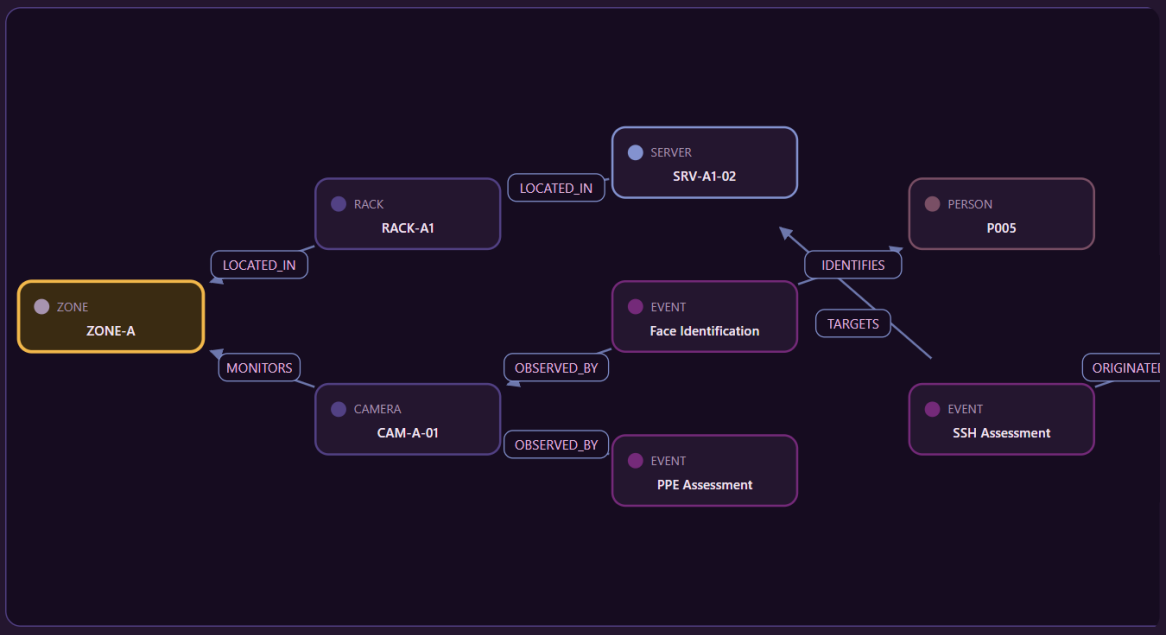
WHY GROUPED	3 events in **ZONE-A**, 0 min apart, linked by 3 shared-context links (same zone and time window).
DOMAINS	Cybersecurity · Physical security · Safety / PPE
WHAT IT DOES NOT SHOW	**Not proof of cause.** Being grouped does not mean one event caused another, or that the same person was involved.
SEVERITY	HIGH set by **fixed rules** (DCG-DECISION-v2), **not by AI**. Raised one level because the events form a known linking mechanism. Group severity rule validated on the author's labels only (no independent labelling round).
HUMAN REVIEW	Required before any action. 0 of 3 events have a human verdict; the group's review state is under Investigation state.

Infrastructure relationships

9 nodes · 8 relationships · Neo4j read-only

+ − Reset

Event · Camera · Person · Server · SourceIP · Zone · Rack Highlighted in amber: the zone where these events were placed



Select a node to inspect properties. Use +/− to zoom; drag the background to pan. Relationships shown are returned by Neo4j, not inferred by the layout.

Select a graph node to view its approved details.

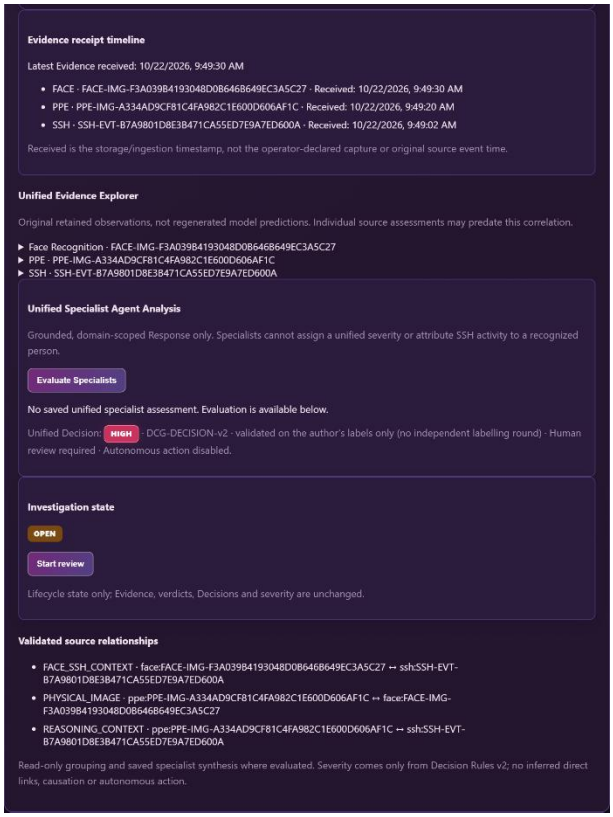


Figure 8. Group detail: the At a glance summary (top) and the Neo4j graph linked to the zone (bottom); Full detail also shows the Decision rules, events and verdicts, and the investigation state

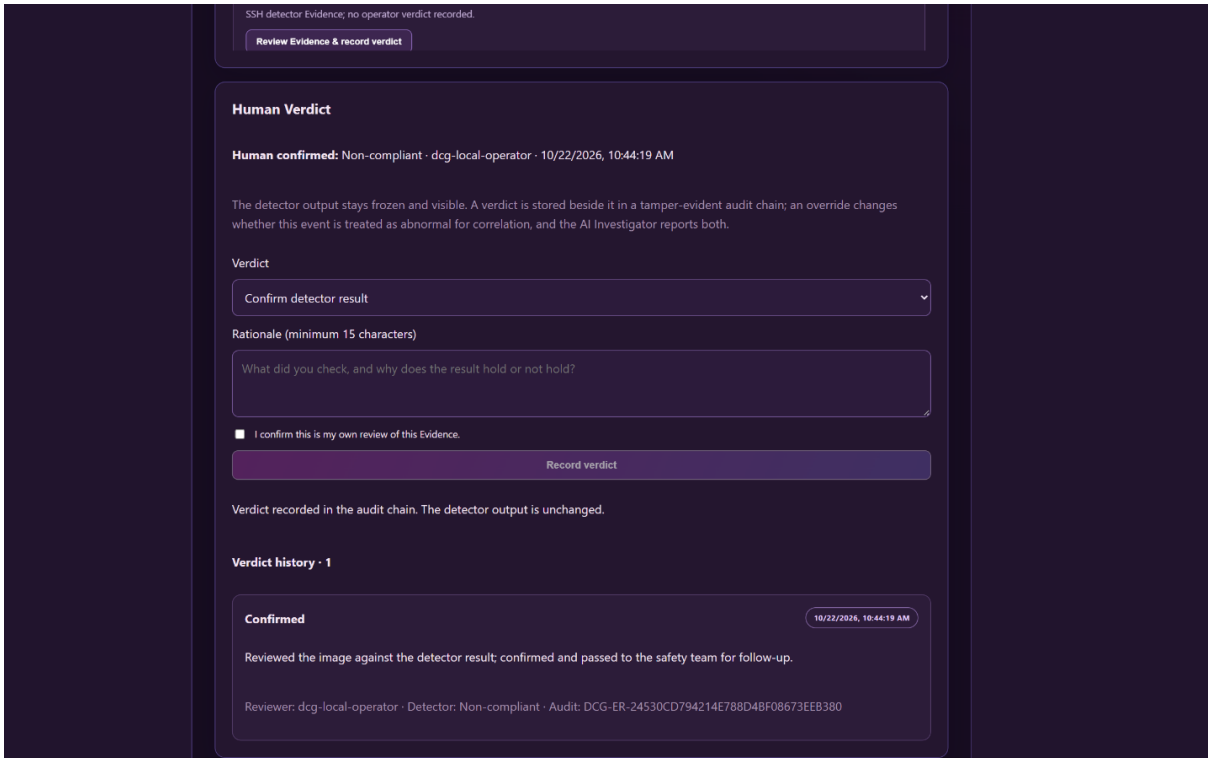


Figure 9. Human Review Queue: a recorded verdict, the audit-chain confirmation and the verdict history

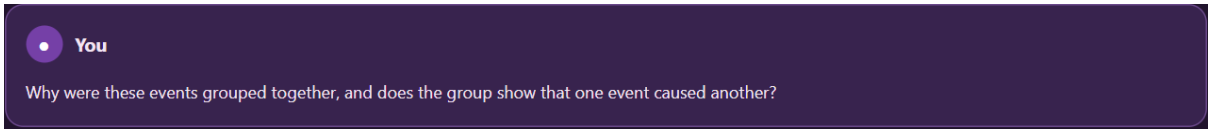


Figure 10. AI Investigator: the operator's question

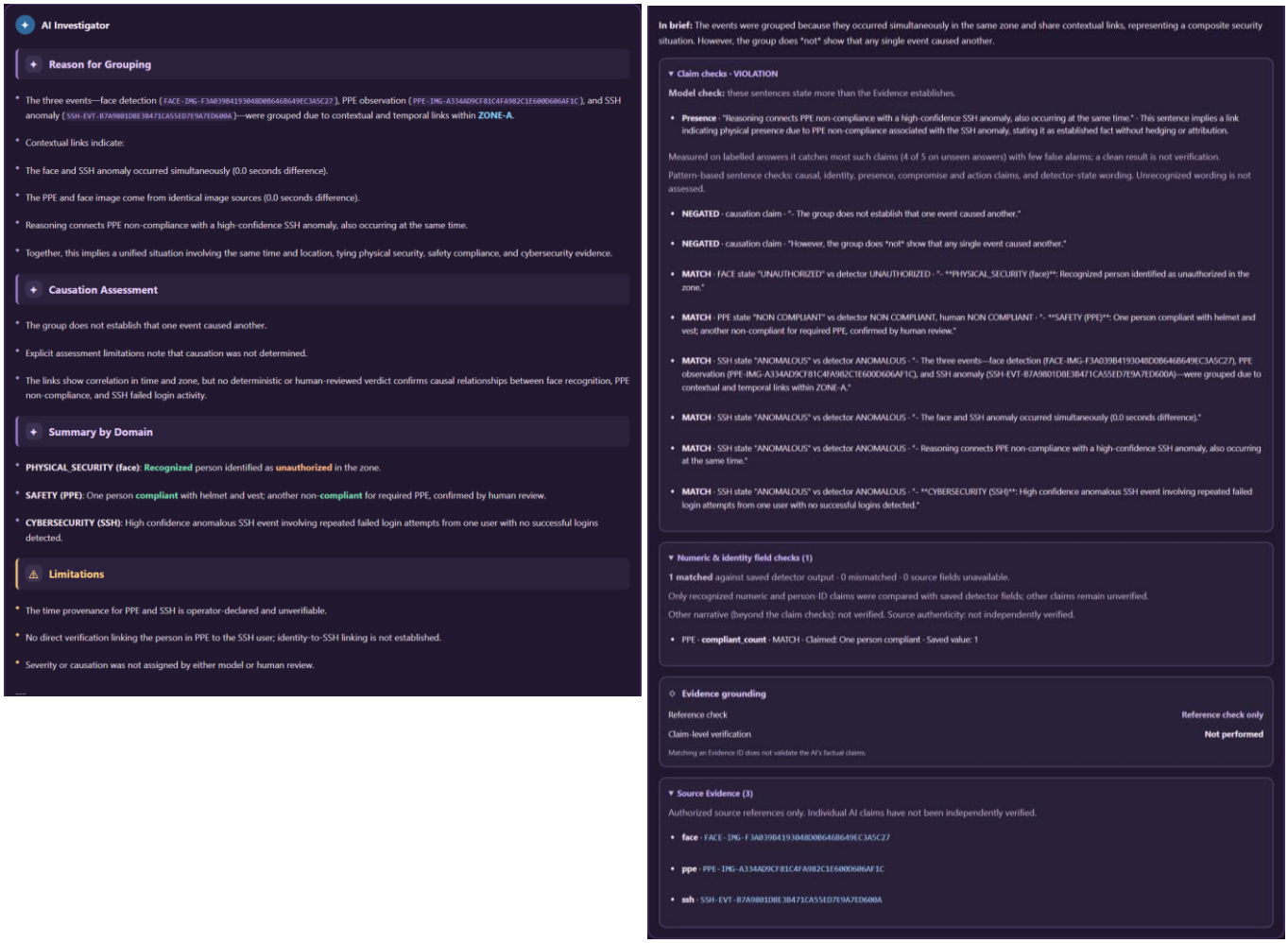


Figure 11. AI Investigator: the full answer, with its claim checks and sources

7.5.1 Architecture of the application

- **Backend:** FastAPI with about 35 routers under `/api/v1`. SQLite stores the presentation state (evidence records, Decisions, verdicts, reviews, audit and optional history); Neo4j is read through read-only graph projections.
- **Frontend:** React 19 and TypeScript, built with Vite. The dashboard refreshes every 5 seconds and shows three times for each record: when it was observed, evaluated and received.
- **Workers:** each detector runs as a persistent worker process, in its own isolated environment when installed, so GPU libraries for PPE and face never mix with the main application; without it, the detector runs on CPU in the main environment.
- **Ingestion:** uploads and an optional watched folder are processed idempotently; raw SSH logs are not retained after processing, while their original timestamps are kept.

7.5.2 Access control

Authorization is deny-by-default and combines a role with the zones a user may see. Roles are never taken from requests; they come from the server-side identity.

Table 25. Roles and what each can see

Role	Can see in addition to incident and graph summaries
Viewer	Nothing further
Security operator	SSH detail
Safety operator	Person and camera detail (face and PPE)
Operations engineer	Maintenance and environmental detail
Administrator	Everything, including scenario execution and audit verification

An automated access-matrix test checks all 99 API operations for every role: requests without credentials get 401, and 403 appears exactly where permission is missing. Evidence images are served only to users authorized for that zone.

7.5.3 Human review, lifecycle and audit

- **Verdicts:** confirm, override or inconclusive on any evidence record in all five domains, with a rationale; detector output is never modified.
- **Group lifecycle (DCG-LIFECYCLE-v1):** open, in review, resolved, with reopen. A group reopens automatically when its evidence or verdicts change. A recorded "needs follow-up" is never treated as a resolution.
- **Audit chains:** four append-only hash-chains (group reviews, evidence verdicts, Investigator calls and the investigation lifecycle). Each record stores its time, policy version, evidence signature and the hash of the previous record, and the reviewer's identity is taken from the server.
- **Tamper detection:** a verifier detects edited records and broken links, and the backend refuses new records if a chain fails. External checkpoints (length and head hash of each chain, optionally signed with an HMAC, kept outside the data folder) also detect deleted records at the end of a chain and a replaced database.

7.5.4 Operations

- **Logging:** request method, path, status and duration, application messages and errors; request bodies, headers, credentials and query strings are never logged.
- **Backups:** a consistent online copy of the database, the images, a graph export and a hash manifest, with a restore check.
- **Security headers and rate limits** are applied, including a limit on outbound model calls.

7.6 Testing approach

- **Every layer is tested with pytest:** 14 suites run by one command (`check_all.py`), including contract tests in dependency order for the Reasoning layer and 24 Playwright browser tests for key flows, including one per usability-study change.
- **Security:** an access-matrix test covers all 99 API operations for every role; continuous integration gates on Bandit static analysis, pip-audit and npm audit.
- **Accessibility:** an automated WCAG 2.1 A/AA scan of every main screen reports zero violations.
- **Continuous integration:** GitHub Actions runs the suites on every push; the main branch accepts changes only through pull requests with passing checks.

7.7 Technical challenges and how they were resolved

Table 26. Technical challenges and how they were resolved

Challenge	Resolution
No dataset spans all five domains in one facility	Real data for each detector; synthetic topology and labelled scenarios for cross-domain reasoning, labelled as such
Uploads took 7 to 18 s because each started a new process and reloaded its model	Persistent inference workers: SSH 7.3 to 0.56 s, PPE 7.6 to 0.60 s, face 17.8 to 3.1 s, with identical results
The PPE detector missed many safety vests	Larger models at higher resolution, external tests on two further datasets, and a PPE-v3 trained with extra data (Section 8.1)
Language models overclaimed, for example implying presence or causation	Field checks, sentence-level claim checks and a post-generation judge that caught 4 of 5 overclaims with 1 false alarm in 35 answers
A model-integrity check failed on Linux because of line-ending conversion	Found by continuous integration; the file is now stored byte-for-byte
Eight scenarios used alerts the environmental detector cannot produce	Scenario set v2 replaced them and was relabeled blind
OpenSSH logs record no year, so the dashboard showed the year 2000	Found in the usability pilot; times now read "year not in log" while stored values stay unchanged

8. Results and Progress

The central result is that correlation outperforms isolated single-domain monitoring on the held-out synthetic scenarios, while each detector's own accuracy is reported separately on its locked test set.

8.1 Detector evaluations (locked test sets, run once)

Table 27. Detector evaluations on locked test sets

Detector	Test data	Key results
SSH anomaly	280 controlled windows (120 normal, 160 attack)	Two-of-three consensus: precision 1.000, recall 0.713, F1 0.832. Broad operational point: recall 1.000, precision 0.678, specificity 0.367
PPE-v1 (YOLOv8n, 640 px)	SH17, 810 images	Precision 0.689, recall 0.524, mAP@0.5 0.545; helmet recall 0.603, vest recall 0.449
PPE-v2 (YOLOv8s, 1,024 px, selected on validation)	SH17, 810 images	mAP@0.5 0.595, helmet recall 0.654, but vest recall fell to 0.245

Detector	Test data	Key results
PPE-v2m (YOLOv8m, runtime, test-informed choice)	SH17, 810 images	mAP@0.5 0.625, vest recall 0.449; compliance status agrees with ground truth on 95.3% of validation images
Face recognition	15 images (6 known, 9 unknown)	6 of 6 identified, 9 of 9 rejected, no false acceptances; median 2.5 s per image
Drive failure risk	2026 Q1 holdout, 17.1 million drive-days, 667 failed drives	ROC-AUC 0.872, PR-AUC 0.033 (133 times the base rate); 323 of 667 failing drives caught; median warning 7 days ahead; precision 3.4% at threshold 0.45
Environmental	Deterministic, no learned model	Validated by contract tests on thresholds and provenance

SSH detector, locked Set B (280 scenarios):

Table 28. SSH detector on locked Set B

Detector	Accuracy	Precision	Recall	F1	False-positive rate
Rules	76.4%	100.0%	58.8%	74.0%	0.0%
Isolation Forest	35.7%	33.3%	12.5%	18.2%	33.3%
Autoencoder	80.0%	79.6%	87.5%	83.3%	30.0%
Hybrid, 1 or more votes	65.7%	64.8%	87.5%	74.5%	63.3%
Hybrid, 2 or more votes	**83.6%**	**100.0%**	**71.3%**	**83.2%**	**0.0%**
Operational detector	72.9%	67.8%	100.0%	80.8%	63.3%

The two-of-three consensus is the cleanest high-confidence point, while the operational detector catches every attack at the cost of many false positives; both are kept as graded evidence states rather than treating every anomaly as an attack.

Drive failure risk, 2026 Q1 holdout (17,064,047 drive-days; 4,285 positive rows; 667 failed drives):

Table 29. Drive failure risk on the 2026 Q1 holdout

Measure	Result
PR-AUC	0.0335 (133 times the 0.025% base rate)
ROC-AUC	0.872
Precision at threshold 0.45	3.4%
Row recall	33.7%
Failing drives warned	323 of 667 (48.4%)
Alerts per 1,000 drive-days	2.5
First warning before failure	median 7 days, mean 5.5 days

PPE detectors, SH17 locked test (810 images):

Table 30. PPE detectors on the SH17 locked test

Model	All-class mAP@0.5	Helmet recall	Helmet mAP@0.5	Vest recall	Vest mAP@0.5
PPE-v1 (YOLOv8n, 640)	0.545	0.603	0.676	0.449	0.372
YOLOv8n, 1,024	0.578	0.592	0.744	0.367	0.383
PPE-v2 (YOLOv8s, 1,024, selected)	0.595	0.654	0.708	0.245	0.317
PPE-v2m (YOLOv8m, 1,024, runtime)	0.625	0.664	0.739	0.449	0.416

The vest results are unstable because the validation and test sets each hold fewer than 50 vests, so a difference of 10 vests can reverse a conclusion.

Generalization of the PPE detector. On an external helmet-and-jacket test set (1,571 images), all models scored well below their SH17 results, and no single model won on both datasets. On a third dataset (CHV), PPE-v3 detected vests best (recall 0.84 against 0.63) and halved false compliance alarms, but missed more real violations (9.3% against 3.2%). The runtime stays on **PPE-v2m** because a missed violation raises no alert, while a false alarm still reaches a human.

8.2 Correlation against isolated monitoring (research question)

Both systems use the same single-event severity policy, so the comparison isolates the effect of correlation. The protocol was fixed before results were computed; the primary result is the held-out test split with the operator's blind labels.

Table 31. Correlation against isolated monitoring (test split)

Measure (24 test scenarios, 18 incidents)	Isolated	DC-GUARDIAN	Difference (95% CI)
Alerts shown	65	44	−21 (−29 to −14)
Alerts with no incident event	25	21	
Incidents shown whole in one alert	0 of 18	14 of 18	
Mean alerts per incident	2.22	1.28	
Share of top-severity alerts that are real incidents	77% (MEDIUM)	100% (HIGH)	
Average precision of the review queue	0.73	0.81	+0.08 (+0.02 to +0.19)
Alerts reviewed until every incident event is seen	64.2	42.2	−22 (−33 to −15)

Every interval excludes zero. With ChatGPT's labels of the same blind pack as ground truth, the direction is the same: 65 to 44 alerts, 8 of 9 incidents whole, top-severity share 50% to 86%, and average precision 0.48 to 0.75.

8.3 Bayesian correlation against the crisp rule

Table 32. Bayesian correlation against the crisp rule

Test split, 59 pairs	Crisp rule F1	Bayesian F1	Difference, 95% CI
Operator's blind labels (25 related pairs)	0.784	0.857	−0.040 to +0.190
ChatGPT labels (13 related pairs)	0.615	0.703	−0.067 to +0.229

The Bayesian model scores higher under both labellings and ranks pairs well without a threshold (average precision 0.92), but the interval includes zero, so its advantage is **not statistically established**. It is therefore shown on the dashboard for information only and does not drive decisions.

8.4 Severity rules

The DCG-DECISION-v2 group rule was selected on the development split and tested once: groups it escalates are genuine incidents far more often than groups it does not (difference 0.51, 95% CI 0.08 to 0.88). These labels came from the scenario author, and the scenarios carry no severity labels, so the test shows that escalation picks out real incidents, not that each level is correct.

8.5 Knowledge grounding and language-model safeguards

- **Retrieval:** on 18 frozen cases, Recall@3 96.1%, Precision@3 71.6%, MRR 0.97; domain filtering and provenance 100% correct.
- **Post-generation judge:** on 40 blind-labelled live answers, it caught 4 of 5 overclaims, with 1 false alarm in 35 correct answers.
- **Prompt injection:** retrieved instructions to escalate severity or claim compromise are not followed (contract test).

8.6 Latency

Table 33. Per-stage latency

Stage	Median
SSH log upload (persistent worker)	0.56 s
PPE image upload (persistent worker)	0.60 s
Face image upload (persistent worker)	3.1 s
Drive-history inference	0.06 s
Neo4j ingestion and correlation of two events	0.05 s
Group severity decision	8 µs
AI Investigator question (86 observed)	3.2 s

Graph reasoning and decisions add well under a second; language-model steps take seconds but run only when the operator asks. Measured on one workstation with one user; no load test was run.

8.7 Usability study

Method. A formative usability study was run on 21-22 October 2026 with three participants, one day after a pilot run by a team member. Before every session the same study data was rebuilt with the project's preparation script, so all participants saw identical incidents. Each participant thought aloud while completing seven core tasks (T1–T7) and, where time allowed, two optional tasks: asking the AI Investigator about the HIGH group (T8) and checking a camera image in the Scenario Lab (T9). A third

optional task (T10, adding sensor readings) was not run. A task scored 1 when completed without help, 0.5 when completed with help or with one wrong detail, and 0 otherwise. After the tasks, participants completed the System Usability Scale (SUS) [46], a six-item true or false check on what the system does and does not claim, and five trust and clarity statements (1 to 5).

Table 34 describes the participants, their experience with monitoring dashboards and their individual SUS scores.

Table 34. Usability study participants and SUS scores

Participant	Background	Monitoring dashboards used before	SUS score
P01	Cybersecurity	Regularly	55.0
P02	IT operations or support	Never	87.5
P03	Data science graduate	Regularly	40.0

A pilot on 20 October found four problems before any participant session. Two were fixed in the interface (SSH log times and the disabled Resolve button), one task card was reworded, and one design choice was kept and will be observed.

Task results. Table 35 summarizes the tasks. Times were recorded for all three participants on T1 and for two of them on the other core tasks; they are rounded to the nearest minute and are indicative only.

Table 35. Usability task results

Task	Participants	Success rate	Help given	Median time
T1 Start of shift: items needing attention, busiest domain	3	100%	0	3 min
T2 Find the most urgent group, its severity and domains	3	83%	1	1 min
T3 Explain why the alerts were grouped	3	50%	0	4 min
T4 Record a verdict on an unreviewed PPE result	3	100%	0	5 min
T5 Start a review of a group and resolve it	3	100%	0	1.5 min
T6 Find work waiting for human review	3	100%	0	1 min
T7 Find the zone with the most evidence	3	100%	0	1 min
T8 (optional) Ask the AI Investigator	2	50%	2	not recorded
T9 (optional) Check a camera image (PPE)	2	50%	2	not recorded

Five of the seven core tasks were completed by every participant without help. This includes T5, resolving an investigation, which the pilot user abandoned after five minutes because the Resolve button stayed disabled without saying why; the explanation added after the pilot removed the problem. The weakest core task was T3: every participant explained the shared zone and time only in part, and it took the longest (about four minutes). The optional tasks T8 and T9 were completed only with the facilitator's help, which shows that the AI Investigator's consent step and the Scenario Lab's separate validation sections need clearer guidance.

System Usability Scale. The mean SUS score was **60.8**, rated "OK" on the adjective scale of Bangor et al. [47] and below the commonly cited average of about 68. The scores varied widely, from 40.0 to 87.5. The participant who had never used a monitoring dashboard gave the highest score, while the two regular dashboard users scored the system lower; one possible reason is that experienced users compare it with more polished commercial tools. With three participants, the mean is an indication rather than a reliable estimate.

What participants understood. Table 36 shows how many participants answered each concept statement correctly; many answers were "not sure". The mean was 2.3 of 6 correct.

Table 36. Concept check: what participants understood

Statement	Correct answer	Answered correctly
Q1: When the system groups two alerts, one causes the other	False	2 of 3
Q2: Severity is set by fixed rules, not by the AI assistant	True	0 of 3
Q3: Recording my verdict overwrites the detector's result	False	1 of 3
Q4: The system can take actions by itself	False	2 of 3
Q5: A non-compliant PPE result proves the equipment was absent	False	1 of 3
Q6: Every alert is meant to be checked by a human first	True	1 of 3

Two of the three participants understood that grouping does not mean causation (Q1) and that the system takes no action by itself (Q4). However, no participant recognized that severity comes from fixed rules (Q2), and only one understood that a human verdict leaves the detector's result unchanged (Q3), that a non-compliant PPE result is not proof of absence (Q5), and that every alert is meant for human review (Q6). These properties are enforced in the system (Section 6.7), but the interface does not yet make them visible enough.

Trust and clarity. Table 37 gives the mean agreement with each statement.

Table 37. Trust and clarity ratings

Statement	Mean (1–5)
R1: I understood why an alert had the severity it had	2.7
R2: I could tell what came from detectors and what came from a human	3.0
R3: The dashboard made clear what the system does not know	3.7
R4: I would trust this system to help me decide what to look at first	4.3
R5: I would trust this system to decide for me without checking (low is desired)	2.0

Participants would trust DC-GUARDIAN to help them prioritize (4.3), which is its intended role, and on average would not let it decide without checking (2.0). One participant, however, agreed that they would trust it to decide without checking, which is a risk of over-trust. The lowest score was for understanding why an alert had its severity (2.7), consistent with the concept check.

Implications. The task flow works: operators could find what needs attention, record verdicts and resolve investigations. The main gap is communication, not function: the rule behind each severity, the difference between detector and human input, and the system's limits need to be shown where the

operator is looking. The following changes were made in response, after the sessions; they have not been re-tested with participants:

1. An At a glance box now opens every group and single event: why the events were grouped, “not proof of cause”, and the severity “set by fixed rules (DCG-DECISION-v2), not by AI”, with the group-severity caveat.

2. Every event shows its review state (“Not reviewed”, “Human confirmed”, “Human override”), a saved verdict is confirmed in a green box that repeats that the detector’s result is unchanged, and a verdict can be recorded from inside a group.

3. Groups state what they do not show; non-compliant PPE results explain that equipment not detected is not proof that it was absent; long explanations are folded into expandable sections, and groups open in a Simple view with Full detail on request.

4. Every AI panel carries a “Who says what” note (the detector produced the result, the AI only explains, fixed rules set severity, the human verdict counts), and a “How to read this screen” panel explains the terms. The Review Queue now has to do, In progress and Completed tabs, and date, zone and type filters were added.

Limitations. This was a small formative study: three participants with cybersecurity, IT operations and data-science backgrounds, not a representative sample of data-center operators, with sessions facilitated by the project team. Times were not recorded for every participant, single-task ease ratings were not collected, and T10 was not run. The changes above were made after the sessions and were not re-tested; repeating T3 and the concept check would measure them. The results show where the interface helps and where it confuses users; they do not support statistical claims about all users.

8.8 Deviations from the initial plan

Design changes from the proposal are listed in Section 6.8. Table 38 covers deviations in what was delivered and evaluated.

Table 38. Deviations in what was delivered and evaluated

Proposal plan	What was delivered	Effect
Optimization of technician assignment	Not built; effort went to model enhancement and faster inference	The search and evolutionary technique group is not covered
Single frozen PPE model	PPE-v2, v2m and v3 trained and tested on three datasets	Better overall detection, with dataset-dependent gains reported honestly
Not in proposal	Bayesian correlation study, usability study, external audit checkpoints	Adds reasoning under uncertainty and a user evaluation
Independent labelling of scenarios	Dropped for time; operator labels with a ChatGPT sensitivity check	Results rest on the author's own labels

8.9 Experiment register

Every experiment was run in the project, what it showed, and which option is in use and why. Selection always used validation or development data; locked tests were run once.

Table 39. Experiment register

#	Experiment	Result	In use, and why
1	SSH: rules, Isolation Forest, autoencoder, two-of-three consensus and a broad operational detector (locked test, 280 windows)	F1: rules 0.740, Isolation Forest 0.182, autoencoder 0.833, consensus 0.832 (precision 1.000), operational recall 1.000	All kept as graded evidence states: consensus gives `HIGH_CONFIDENCE_ANOMALY`, one vote gives `ANOMALY_CANDIDATE`. Isolation Forest is only a novelty vote because it is unreliable alone
2	SSH external behavioral check on a honeypot dataset (145,425 events) [29]	Confirmed that source-IP windows, root targeting, username diversity and bursts matter; scanning without login is not covered	Not used for accuracy; informs future features
3	Drive risk: SMART baseline, plus health indicators, Temporal Random Forest, HistGradientBoosting (2025 Q4 validation)	PR-AUC 0.018, 0.022, 0.026, 0.024	**Temporal Random Forest v2:** best PR-AUC, the primary measure under extreme class imbalance
4	Drive risk: tree settings and negative-sampling ratios	50 negatives per positive improved validation PR-AUC	50:1 sampling in the frozen model
5	Drive risk: GRU sequence model against the Random Forest on identical Q4 endpoints	PR-AUC 0.016 against 0.023	Random Forest kept; GRU reported as a challenger
6	Drive risk: operating threshold curve (validation)	0.45 sits at a knee; 0.30 catches 60% of failing drives instead of 50% but flags 2.3 times as many	**0.45** (operator decision): fewer false alerts; 0.30 documented as the alternative
7	Face: distance threshold on validation (7 known, 6 unknown)	Known distances at most 0.33, unknown at least 0.79	**0.50**, in the middle of the gap; locked test 15 of 15 correct
8	PPE-v1: YOLOv8n at 640 px on SH17	mAP@0.5 0.545; vest recall 0.449	Baseline; still selectable
9	PPE-v2: YOLOv8n, s and m at 1,024 px, selected on validation by a pre-declared rule	Validation: 0.616, 0.648, 0.648; YOLOv8s chosen on the tie-break, then lost vest recall on the locked test (0.245)	Official PPE-v2, not used at runtime

#	Experiment	Result	In use, and why
10	PPE: post-hoc locked-test scores of the unselected candidates	YOLOv8m best: mAP@0.5 0.625, vest recall 0.449	**PPE-v2m at runtime** (operator decision, recorded as test-informed); compliance agreement on validation 95.3% against 94.8%
11	PPE: external test on Safety Helmet and Reflective Jacket (1,571 images)	All models far below their SH17 scores; v1 best on helmets, v2m best on vests	Gains reported as dataset-dependent
12	PPE-v3: SH17 plus the Kaggle training data, persons pseudo-labelled	Kaggle helmet and vest mAP@0.5 0.962 and 0.958; on SH17 no better than v2m (0.619 against 0.625)	Not deployed; best detector on new sources
13	PPE: third-dataset compliance test on CHV	v3m: vest recall 0.84 against 0.63, false alarms 31.5% against 59.9%, but missed violations 9.3% against 3.2%	**Runtime stays PPE-v2m:** a missed violation raises no alert, while a false alarm still reaches a human
14	Inference: one process per upload against persistent workers	SSH 7.3 to 0.56 s, PPE 7.6 to 0.60 s, face 17.8 to 3.1 s, identical results	**Persistent workers**
15	RAG ranking: raw cosine, maximum diversity, controlled diversity, controlled plus cosine abstention	Recall@3 84.3%, 90.2%, 90.2%, 49.0%; Precision@3 74.5%, 47.1%, 65.7%, 31.4%	**Controlled diversity, top-3:** best balance. Cosine abstention rejected for false abstention; a deterministic eligibility check is used instead
16	RAG re-run after trimming OSHA documents to their own sections	Recall@3 90.2% to 96.1%; MRR 0.87 to 0.97	Trimmed corpus
17	Sentence-level claim checker on held-out answers	Caught 0 of 3 overclaims; no false alarms	Kept as a cheap check, not relied on alone
18	Tighter instructions to the language model (blind before and after)	Overclaims 2 of 20 against 3 of 20: no measurable effect	Not relied on
19	Post-generation judge (40 blind-labelled live answers)	Caught 4 of 5 overclaims; 1 false alarm in 35	**In use**
20	Bayesian network against the crisp rule, with ablation and perturbation (three labellings)	Bayesian F1 higher under all three; significant only on the original v1 author labels	**Crisp rule drives decisions;** Bayesian shown for information because its advantage is not established on the operator's labels

#	Experiment	Result	In use, and why
21	Group severity: four linkage rules compared with the development split	Known mechanism within 10 minutes best; on the test split escalated groups are far more often genuine (difference 0.51)	**DCG-DECISION-v2 with rule L2**
22	Correlation against isolated single-domain monitoring	65 to 44 alerts; average precision 0.73 to 0.81	Answers the research question (Section 8.2)
23	Per-stage latency benchmark and access logs	Reasoning and decisions under 1 s; detectors dominate	Informed the persistent-worker change
24	Usability as a pilot and study	Pilot found 4 problems, 3 fixed; study with 3 participants: 5 of 7 core tasks completed by all without help, SUS 60.8, concept check 2.3 of 6	Task flow works; severity rules and system limits were not seen. The dashboard was changed in response, not yet re-tested (Section 8.7)

9. Challenges

9.1 Project Challenges

The obstacles concern evidence and deployment, not missing features: the results rest on synthetic scenarios and the author's own labels (no independent labelling round), and the prototype is not yet built for a real facility.

Data

- No synchronized real-world dataset spans all five domains, so cross-domain scenarios rely on synthetic topology and controlled timing.
- Synthetic mappings are evaluation constructs, not observations from a real data center.
- Large, private, biometric, or external datasets are excluded from Git: raw Backblaze, SH17, and SSH datasets, face photographs, and face enrollment embeddings.
- Credentials and `.env` files must never be committed.
- Checks that need private local data are skipped when that data is absent.

*Model and Evaluation C**hallenges***

Table 40. Model and evaluation limitations by component

Component	Key limitation
Face Recognition	Locked test used only 15 images (6 known, 9 unknown); 100% results are proof-of-concept and not generalizable biometric accuracy.
PPE Detection	Modest recall (runtime PPE-v2m: helmets 0.664, safety vests 0.449). A NON_COMPLIANT result means PPE was not detected, not that it is physically absent.

Component	Key limitation
Predictive Maintenance	Extremely rare positives (about 0.025%) give low precision (about 3.4%); 323 of 667 failed drives detected on the final holdout.
SSH Anomaly Detection	The operational detector reaches full recall but low specificity (0.367); the stricter hybrid consensus has zero false positives but lower recall (0.713). Metrics come from controlled scenarios, not real attack prevalence.
Environmental Monitoring	Not a learned model, so no precision or recall is reported; thresholds must eventually match real sensor and equipment specifications.

Metrics from different components use different protocols and must not be compared as if from one benchmark.

Architectural Challenges

- Evidence relationships are established through graph and explicit deterministic rules before any LLM reasoning.
- Source-model assessments and provenance are preserved, not rewritten by later layers.
- RAG retrieves approved knowledge only and cannot decide or revise correlation.
- The Decision layer is rule-based and deterministic.
- Executable Reasoning changes must keep all 24 of 24 contracts passing, unless the suite is intentionally versioned for a documented interface change.

Technical and Environmental Constraints

- Python 3.12 with the repository-level `requirements.txt` (a curated direct-dependency list, not a full freeze).
- Neo4j must be running for graph and correlation contracts.
- Git LFS is required to retrieve the frozen Temporal Random Forest model.
- The setup script (`setup.ps1`) targets Windows; no GPU is needed for the demo, but GPU tests run in separate environments.
- The AI Investigator, the specialists, and the Decisions for SSH events and drive + room-temperature pairs require an OpenAI key; the automated checks do not call OpenAI.

Security and Governance Challenges

- Security results (0 High, 0 Medium, 5 reviewed Low Bandit findings; no known vulnerabilities from pip-audit at submission, 23 October 2026) are not a penetration test or certification.
- The AI Investigator is opt-in and read-only.
- Human review actions are authenticated and audited.

Table 41. Remaining challenges and strategies

Remaining challenge	Implication	Strategy
Synthetic data and topology	Results describe designed scenarios, not real incidents	Pilot on a real site's historical logs and alarms with operator consent
Bayesian advantage not established	It cannot yet drive decisions	Model slow cooling cascades by conditioning time on topology; re-test on independent labels

Remaining challenge	Implication	Strategy
PPE generalization	Accuracy drops on new image sources; vests remain the weak class	Collect data-center images with consented staff; re-tune PPE-v3 on validation data
Small face test (15 images)	Biometric accuracy and fairness are not established	Larger, consented evaluation before any real use
Licensing	SH17 is non-commercial, YOLOv8 is AGPL-3.0, and the face weights are research-only	Commercially licensed data and models before release
Single-user prototype	HTTP Basic login, SQLite and one workstation are not production controls	Identity provider, PostgreSQL, containers behind TLS, and an independent security review
Correlation misread as cause (usability concept check 2.3 of 6)	Operators may treat linked events as proof of cause and over-trust them	Re-test the post-study changes (summaries, wording, help panel) with task T3 and the concept check

9.2 Strategy for Improvement

This strategy addresses seven constraints, C1 to C7, listed with their actions in the Challenge Strategy Summary at the end of this section. It proceeds in four phases: strengthening the evidence base, improving the weakest models, assessing the value of each reasoning technique, and preparing for deployment.

Phase 1: Strengthen Validation (C1, C2, C3)

- Build a larger library of labeled cross-domain scenarios (for example, a cooling fault combined with an SSH anomaly and a drive-risk alert) with ground-truth correlations, and report correlation precision and recall against it.
- Add fault-injection scenarios for rare events and require the system to abstain and escalate to human review when a case is unseen.
- Create additional synthetic topologies (for example, DC-02 and DC-03) with different sizes and layouts, and test topology changes such as a rack being moved or a service being migrated.
- Seek anonymized logs or sensor data from a real or partner facility, even a small sample, to test how well synthetic results transfer.
- Have independent practitioners label the scenarios, and re-run the correlation, severity and Bayesian evaluations against their labels.

Phase 2: Improve the Weakest Models (C4)

- Collect data-center PPE images with consented staff to choose between PPE-v2m and PPE-v3; the actions for each component are listed below.

Table 42. Improvement actions by component

Component	Improvement actions
PPE Detection	Rebalance classes, add data augmentation and data-center images, and tune per-class thresholds, starting with the safety vest class.
Predictive Maintenance	Tune the decision threshold and evaluate precision at top-k, since low precision is driven by extreme rarity of failures.
SSH Anomaly Detection	Use the stricter consensus mode where false positives are costly, and add more benign traffic to raise specificity.
Face Recognition	Expand the test set to more subjects and report false accept and false reject rates.
Environmental Monitoring	Calibrate thresholds to real equipment and sensor specifications.

Phase 3: Assess Reasoning Quality (C7)

- Extend the ablations, so far run for the Bayesian network (Section 8.3), to the whole pipeline (graph only, graph plus RAG, full pipeline) to show what each reasoning technique contributes.
- Measure RAG grounding quality, including abstention when knowledge is insufficient.
- If root-cause support is desired, present it as ranked candidate hypotheses for operator review, not as conclusions.

Phase 4: Deployment Readiness (C5, C6)

- Containerize the stack and run load and integration tests.
- Run a shadow-mode pilot in which the system advises but takes no action.
- Commission an independent security review, such as a penetration test.
- Repeat the usability study with data-center operators, re-testing the changes made after the first study, and use their review decisions to refine decision rules.

Challenge Strategy Summary

Table 43. Challenge strategy summary (C1 to C7)

Constraint	Primary actions	Evidence of improvement
C1 Limited multimodal data	Scenario library, partner data	Correlation precision and recall on labeled scenarios
C2 Rare-event generalization	Fault injection, abstention and escalation	Correct escalation rate on unseen scenarios
C3 Controlled topology	Additional topologies, topology-change tests	Stable results across topologies
C4 Model performance	Rebalancing, threshold tuning, larger test sets	Improved per-class recall, precision, and specificity
C5 Deployment and resources	Containerization, load tests, security review	Pilot completed and review findings resolved

Constraint	Primary actions	Evidence of improvement
C6 Human in the loop	Shadow-mode pilot, operator study	Operator agreement with recommendations
C7 Contextual, not causal	Ablations, hypothesis ranking	Measured contribution of each technique

10. Future Work

Although DC-GUARDIAN demonstrates the potential of integrating multiple monitoring domains through intelligent reasoning, further development is needed to strengthen its reliability, scalability, and readiness for real-world data center operations. The following areas represent the main priorities for future improvement.

- **Independent validation:** Practitioner labelling of the scenarios and a pilot on real historical data remain the strongest next steps for both research and market credibility. The current evaluation relies on controlled scenarios and project-defined labels, which may not fully represent real operational conditions. Future work should involve experienced cybersecurity analysts, data center operators, and maintenance engineers in independently reviewing and labelling incidents. A pilot using historical security logs, equipment telemetry, physical access records, and environmental data would help assess correlation accuracy, incident prioritization, false-positive rates, and the usefulness of generated recommendations. These activities would strengthen the reliability of the findings and provide evidence of the system's practical value.
- **Stronger uncertainty reasoning:** A Bayesian network in which time depends on topology, so slow cooling cascades are linked. The current Bayesian model estimates relationships between events using temporal proximity, infrastructure topology, domain mechanisms, and evidence strength. However, some incidents develop gradually and may not fit within fixed correlation windows. Future improvements could introduce more advanced probabilistic dependencies between event timing and infrastructure relationships. For example, a cooling system malfunction may gradually increase server temperatures and contribute to hard-drive degradation over several hours. Modeling these delayed relationships would improve the identification of developing incidents, support more accurate uncertainty estimates, and distinguish meaningful operational connections from coincidental events.
- **Resource optimization:** Technician dispatch and maintenance scheduling with informed search or a genetic algorithm, using the knowledge graph for locations and skills. Future development could extend DC-GUARDIAN beyond detecting equipment risks to recommending efficient maintenance plans. Optimization algorithms could prioritize maintenance tasks based on predicted failure risk, equipment criticality, technician availability, required expertise, and operational constraints. The Neo4j knowledge graph could provide information about asset locations, equipment dependencies, and technician responsibilities. This would support better resource allocation, reduce maintenance delays, and enable proactive intervention before potential hardware failures affect service availability, while keeping final decisions under human supervision.
- **Live integration:** Streaming connectors for syslog, SIEM, video management and DCIM or BMS instead of uploads. The current prototype primarily processes controlled data uploads, which support reproducible evaluation but limit continuous monitoring. Future work should establish real-time connections with existing cybersecurity, physical security, equipment-health, and environmental monitoring systems. Standardized APIs and event-streaming technologies could enable continuous data collection, automated processing, and timely cross-

domain correlation. This would improve situational awareness, support earlier identification of developing incidents, and reduce dependence on manual data uploads.

- **Production hardening:** Single sign-on, encryption in transit, production database and object storage, monitoring, backups, and penetration testing. Moving DC-GUARDIAN toward operational deployment requires stronger security controls, infrastructure resilience, and scalability. Future enhancements should include centralized authentication, secure credential management, strengthened access controls, and encrypted communication between system components. Production-grade databases, secure object storage, automated backups, and disaster-recovery procedures would improve reliability and maintainability. Comprehensive monitoring, vulnerability assessments, penetration testing, and performance testing would also be necessary to verify that the system can operate securely under realistic workloads.
- **Alert-fatigue controls:** Suppression and deduplication tuned with operators. Although DC-GUARDIAN demonstrated improvements in reducing fragmented alerts through cross-domain correlation, additional mechanisms could further minimize unnecessary notifications. Future development should introduce configurable suppression rules, event deduplication, and grouping of repeated observations related to the same unresolved incident. For example, consecutive failure-risk predictions for one hard drive could be consolidated into a single maintenance alert rather than generating repeated notifications. These mechanisms should consider event persistence, detector confidence, asset criticality, and operator feedback to reduce alert overload without suppressing important warnings.
- Work assignment and onboarding: assign review items to named people or teams, show what each team is working on, filter the queue by user or team, and add an interactive first-time tour, as requested in the usability study (Section 8.7).

Overall, these future enhancements would build upon DC-GUARDIAN's established capabilities by expanding its functionality, scalability, and applicability to more complex data center environments. By advancing contextual reasoning, maintenance optimization, real-time integration, and operational efficiency, the system could further strengthen its role as a comprehensive intelligent monitoring solution. These developments would extend its existing capabilities while maintaining its core principles of explainability, evidence traceability, transparent decision-making, and human oversight.

11. Conclusion

DC-GUARDIAN demonstrates that a governed reasoning layer can transform fragmented data-center alerts into fewer, better-explained incidents while preserving human responsibility for decisions. By connecting evidence across operational, physical security, and cybersecurity domains, the system provides a more coherent view of events and helps analysts prioritize their attention. Its contribution lies in combining complementary AI and reasoning techniques within a framework that makes their outputs traceable and subject to oversight.

- **Academic contribution:** The project presents an integrated pipeline combining machine learning detectors, a knowledge graph, deterministic rules, Bayesian reasoning, and grounded language models, with tested interfaces between components. Evaluation on held-out scenarios supports the research hypothesis: a 32% reduction in alerts, more complete incident views for most incidents, and improved prioritization of the review queue, with reported intervals excluding zero. These findings provide evidence that combining detection with contextual reasoning can improve incident consolidation and prioritization under the evaluated conditions.
- **Market relevance:** The solution addresses a gap between DCIM, SIEM, video surveillance, and asset management tools by enabling reasoning across these systems. It offers a way to connect their outputs and interpret related events within a shared operational context.

Auditability, access control, and human oversight are incorporated into its design, supporting the need for accountable and explainable decision support in data-center operations.

- **Bridge to practice:** The project clearly identifies what remains unvalidated, including performance on real-site data, evaluation against independently labelled incidents, and the adequacy of production controls. These limitations define a practical roadmap for progressing from a research prototype to a pilot deployment. The next stage should assess whether the observed benefits hold under realistic workloads, diverse incident patterns, and operational constraints, alongside feedback from analysts who would use the system.

Taken together, the results establish a foundation for further validation of DC-GUARDIAN as an integrated decision-support approach. Its practical value will depend on demonstrating that reduced alert volume and clearer incident explanations translate into consistent improvements in operational review.

Supplementary Materials

Appendix A. Project Proposal

The original proposal, *DC-Guardian: Data Center Physical & Cybersecurity Correlated with Predictive Maintenance*, is attached as a separate document. Section 6.8 explains how the design changed. Its key commitments and their status:

Table 44. Proposal commitments and their status

Proposal item	Status at submission
Detectors for physical security (face, PPE), cybersecurity (SSH) and equipment health, with outputs in a common event format	Delivered and extended to five domains, each detector evaluated on a locked test
Neo4j knowledge graph of a synthetic data center	Delivered, with graph-derived authorization
A correlation agent to decide which events form one incident	Replaced by deterministic correlation rules, with a Bayesian probability shown as context
RAG knowledge base of SOPs, policies and procedures	Delivered and evaluated over 13 approved documents
Specialized agents that assess incidents and recommend responses	Delivered as grounded specialists that explain evidence with checked citations; they do not set severity
Rules for severity and escalation	Delivered (DCG-DECISION-v1 and v2)
Streamlit dashboard with a chat interface	Delivered as a FastAPI and React application with an opt-in, read-only AI Investigator
Optimization of technician assignment or maintenance scheduling	Not delivered; future work
Research question: improvement over isolated single-domain monitoring	Answered (Section 8.2)

Appendix B. Mapping of System Functions to Course Techniques

Table 45. Mapping of system functions to course techniques

System function	Technique	Course
Correlation rules: abnormal, different domains, same zone, within 15 minutes	Rule-based reasoning; knowledge-based reasoning	MR, RS
DCG-DECISION-v1 and v2 severity rules, with explanation of the rules triggered	Business rules and decision automation; explanation facility	RS
Neo4j model of zones, racks, servers, cameras, sensors, people and authorizations	Knowledge representation; knowledge graph	MR, CGS
Authorization derived from `AUTHORIZED_FOR` paths	Reasoning over a knowledge graph	MR
Bayesian network for P(Related), with likelihood-ratio explanations	Reasoning under uncertainty; probabilistic reasoning	MR
Expert knowledge base for the Bayesian network, with a source for every number	Knowledge acquisition	MR, RS
Drive failure risk with a Temporal Random Forest; GRU challenger	Supervised learning; representation of models from machine learning	MR, RS
SSH Isolation Forest and autoencoder	Unsupervised anomaly detection; neural networks	MR
PPE detection with YOLOv8	Vision cognition; deep learning	CGS
Face recognition with RetinaFace and ArcFace	Vision cognition; similarity matching on embeddings	CGS
RAG over approved documents with local embeddings	Natural-language cognition; retrieval and similarity-based reasoning	CGS, RS
Specialist agents, synthesis and AI Investigator	LLM reasoning; generative AI project lifecycle	CGS, MR
Claim checks and post-generation judge	Guardrails for LLM reasoning	CGS
Scenario evaluation with bootstrap intervals, ablation and sensitivity analysis	Evaluation of reasoning systems; knowledge discovery from data	RS

MR: Machine Reasoning. RS: Reasoning Systems. CGS: Cognitive Systems.

Appendix C. Installation and User Guide

C.1 Requirements

- Windows 10 or 11 (tested on Windows 11), 16 GB RAM recommended (8 GB works for the demo; setup and every check passed on a 7.4 GB laptop) and 15 GB free disk; Python 3.12; Node.js 20 or later; Git with Git LFS. After installing them, close and reopen PowerShell, then check that git --version, git lfs version, py -3.12 --version and node --version each print a version.

- Neo4j (tested with 2026.08, for example in Neo4j Desktop), started and running at bolt://127.0.0.1:7687. The instance name can be anything; the database user is always neo4j.
- No GPU is needed. Optional: an NVIDIA GPU for faster PPE checks (.venv-ppe), and an OpenAI API key for the AI Investigator, the specialists, and the Decisions for SSH events and drive + room-temperature pairs, whose rules need a specialist analysis first. Without a key, correlated groups and single PPE, face, drive and sensor results still get a rule-based severity.

C.2 Installation

```
git lfs install
git clone https://github.com/Jowharah/IRS-PM-2026-10-25-AIS-GRP22-DC-Guardian-System.git
cd IRS-PM-2026-10-25-AIS-GRP22-DC-Guardian-System\SystemCode
git lfs pull
powershell -ExecutionPolicy Bypass -File setup.ps1
```

setup.ps1 asks for the Neo4j password and a new dashboard password. It then creates the Python environment, writes .env (login user DCG_LOCAL_USER, operator by default; AI Investigator off), installs the frontend, loads the DC-01 topology into Neo4j, builds the knowledge index from the 13 bundled documents and loads demo data. It takes 15 to 40 minutes, mostly package downloads (15 minutes on the test laptop), and is safe to re-run. For an unattended run, pass -NeoPassword and -DashboardPassword.

C.3 Running the system

```
.venv-dc-guardian\Scripts\python.exe -m uvicorn presentation.backend.app.main:app --host 127.0.0.1 --port 8000
cd presentation\frontend
npm run dev
```

Run the two commands in two terminals inside SystemCode. Open http://127.0.0.1:5173 and sign in with the user in DCG_LOCAL_USER in .env (operator unless changed) and the dashboard password chosen during setup.

C.4 Verifying the installation

```
.venv-dc-guardian\Scripts\python.exe check_all.py
```

It runs every test suite (about 5 minutes) and writes logs\checks\<time>\report.md. GPU suites and tests that need private data (face photos, training datasets) are skipped with a reason; the live API check runs while the backend is running. The checks never call OpenAI, so passing checks do not show that the Decisions needing a key exist. On the test laptop: about 5 minutes, 12 passed, 0 failed, 2 skipped (GPU environments). The full guide, with troubleshooting, is SystemCode/docs/INSTALLATION_AND_USER_GUIDE.md (also ProjectReport/DC-Guardian_Installation_and_User_Guide.pdf).

C.5 Using the dashboard

1. **Scenario Lab:** upload an SSH log, a SMART CSV, a sensor CSV, a PPE image or a face image, and declare the camera and capture time where asked.
2. **Monitoring Center: see what needs attention (How to read this screen explains the terms), filter the feed by date, zone and severity, and open any evidence record or correlated group: it starts with At a glance, and groups offer a Simple view and Full detail with the graph.**
3. **Human Review Queue: work through To do, In progress and Completed; confirm, override or mark evidence as inconclusive, and move correlated groups through open, in review and resolved, with a written rationale.**
4. **Analytics: choose a period or custom dates, see review progress, filter by domain, severity and zone, and click through to the records.**

5. **AI Investigator (optional):** select evidence or a group, consent to the request, and ask questions. Answers list their sources and flag unsupported claims.

C.6 Data not included

Face photographs, enrolment embeddings, raw datasets and credentials are not in the repository for privacy and licensing reasons. The face workflow needs a local enrolment set; all other workflows run on the controlled inputs provided and set up loads demo data (an SSH burst in Zone A, drive and sensor readings in Zone B).

Appendix D. Individual Project Reports

D.1 AL AJMI ALJAWHARAH MOHAMMAD H (A0356913X)

1. My contributions to the project

As group leader and technical integration lead, I coordinated task allocation, system architecture, and development across all five monitoring domains. I established the unified GitHub repository, managed continuous integration, converted colleagues' Jupyter Notebooks into Python modules, and hosted the integrated prototype on my workstation. I developed the SSH cybersecurity anomaly-detection component using rules, Isolation Forest, and an autoencoder, applying cybersecurity expertise to behavioral feature engineering. I contributed to the Common Event Schema, cross-domain correlation, and Neo4j knowledge graph integration. I oversaw FastAPI backend and React frontend design, RAG and AI agent integration, human review, and auditability. As quality-assurance lead, I coordinated end-to-end testing, security and dependency vulnerability checks, troubleshooting, and performance optimization. I also participated in usability testing and contributed to report Sections 5–8.

2. Self-reflections

DC-GUARDIAN allowed me to expand my existing AI and cybersecurity engineering experience into intelligent reasoning systems. My most significant learning was knowledge representation using Neo4j, where relationships between people, assets, locations, and events provide context beyond individual model predictions. I gained practical understanding of combining symbolic rules, Bayesian reasoning, and grounded generative AI while maintaining explainability, security, and human oversight. Leading multidisciplinary development strengthened my architectural decision-making, technical coordination, and quality-assurance skills. I also learned the importance of integrating independently developed models through consistent interfaces and rigorous validation. In my professional field of AI for cybersecurity, I can apply these concepts to correlate fragmented security alerts, enrich investigations with contextual knowledge, and develop trustworthy AI-assisted security operations solutions. This experience reinforced my ability to lead complex industrial AI projects that connect technical innovation with practical operational needs.

D.2 ALOTAIBI MARAM MUGAAD M (A0355452A)

1. Personal Contribution to the Group Project

My contributions to the DC-GUARDIAN project covered both technical development and research. I was responsible for developing the PPE Compliance Detection module using YOLOv8 to identify personal protective equipment and assess worker compliance in data-center environments. I also contributed to the development of the Retrieval-Augmented Generation (RAG) system and customized AI agents to support context-aware reasoning, evidence interpretation, and intelligent responses to detected events. Additionally, I was responsible for writing the literature review and conducting market research for the project proposal. This involved reviewing relevant academic studies, examining existing solutions, identifying research gaps, and highlighting DC-GUARDIAN's potential

contributions. Through these responsibilities, I contributed to both the project's technical implementation and its research foundation.

2. Self-Reflection

This project helped me understand the difference between developing an AI model and building a component that contributes to a larger intelligent system. Working on DC-GUARDIAN provided me with valuable experience in applying artificial intelligence to real-world security and operational challenges. Developing the PPE Detection module strengthened my understanding of computer vision, object detection, model training, and evaluation. Working with RAG and customized AI agents expanded my knowledge of how large language models can retrieve relevant information, interpret evidence, and support intelligent decision-making. Writing literature reviews and conducting market research also improved my research, analytical, and critical-thinking skills, particularly in identifying limitations in existing solutions and opportunities for innovation. One of the main challenges was understanding how different AI components could work together within a unified system. Overall, this project enhanced my technical confidence, problem-solving abilities, and appreciation of the importance of combining academic research with practical AI development.

D.3 ALSHAKHS MAI ZAID M (A0354315J)

1. Personal Contribution to the Group Project

I conducted three literature reviews on cyber-physical systems to identify existing approaches and unexplored solutions, which led to defining the project scope, outlining five evidence domains and six reasoning techniques. I developed the facial recognition component, pairing RetinaFace detection with ArcFace embedding to supply robust identity evidence for physical-security reasoning, achieving 100% accuracy on a controlled locked test of 15 images (a proof of concept, not general biometric accuracy). Furthermore, I evaluated key project constraints and supported the development of the backend API and frontend auditability workflows, assisting in connecting RAG and configuring specialized AI agents. These efforts connected isolated detections to grounded explanations, governance, and traceable decisions, turning standalone models into a cohesive decision-support system. Finally, I addressed the project limitations and challenges and provided the strategy forward, by employing a fixed, controlled test set and restricted performance claims to proof-of-concept validation. To protect biometric privacy, facial images and enrollment embeddings were excluded from version control. Furthermore, I evaluated known and unknown individuals separately to assess recognition and rejection capabilities.

2. Self-Reflection

This project significantly enhanced my technical, analytical, and communication competencies. Developing the facial recognition component strengthened my ability to translate theoretical research into functional, testable implementations. Evaluating project constraints also reinforced the importance of critically interpreting experimental results and acknowledging limitations to maintain research credibility.

A key challenge involved balancing the project's technical ambitions with the available evidence. I addressed this by presenting findings as proof-of-concept results. Additionally, I established clear distinctions between the prototype's contextual correlation capabilities and causal root-cause determination, which remains beyond its scope. Understanding the integration of facial recognition within the broader reasoning pipeline further strengthened my systems-level perspective. Future

improvements would include expanding the evaluation dataset and conducting tests under realistic environmental conditions. Overall, I recognize the importance of systematic documentation, human oversight for new, unseen situations, and evidence-based decision-making throughout project development.

D.4 ALSHAMMARY NORAH MOHAMMED H (A0354356Y)

1. Personal Contribution to the Group Project

My contribution to DC-GUARDIAN focused on developing the Predictive Maintenance and Equipment Health component for early hardware failure detection and improved data center reliability. Using the Backblaze Drive Stats dataset, I developed a hard-drive failure prediction pipeline, addressing failure labeling, censored observations, class imbalance, and temporal feature engineering. I evaluated Isolation Forest, Random Forest, and GRU models while investigating sampling strategies and decision thresholds. The final Temporal Random Forest achieved a ROC-AUC of 0.872, identifying 323 of 667 failing drives. I also contributed structured maintenance-risk assessments, exploring integration with Neo4j knowledge graphs, Bayesian reasoning, Retrieval-Augmented Generation, and specialized AI agents. This included enabling the Operations Agent to retrieve approved procedures and provide evidence-grounded recommendations while preserving detector outputs and human oversight. Additionally, I contributed to writing the project's Introduction and market-context sections covering reliability challenges, fragmented monitoring, and integration across predictive maintenance, physical security, cybersecurity, and intelligent reasoning.

2. Self-Reflection

DC-GUARDIAN allowed me to strengthen my machine learning and data engineering skills while deepening my understanding of intelligent reasoning in data center operations. Developing a predictive maintenance model using real-world SMART telemetry taught me to address rare failures, temporal dependencies, and incomplete observations. I gained practical experience in feature engineering, chronological validation, class imbalance management, and comparing supervised and unsupervised approaches. Evaluating Random Forest against a GRU sequence model strengthened my model selection skills, while threshold optimization highlighted tradeoffs between early warnings and manageable false-alert rates. I also learned how knowledge graphs connect predictions with asset relationships, environmental evidence, and contextual reasoning. Writing the Introduction and Project Background improved my ability to connect technical development with industry needs. This experience reinforced the importance of reproducibility, explainability, rigorous evaluation, and human oversight while strengthening my ability to translate machine learning into practical solutions supporting proactive maintenance, operational continuity, and informed decisions.

Appendix E. Generative AI Use Disclosure

Table 46. Generative AI use

Tool	Purpose	How its output was checked
ChatGPT (OpenAI) and ChatGPT image generation	Proposal stage: market and literature research, drafting, the proposal architecture image (see the proposal's own disclosure)	Reviewed and edited by the team

Tool	Purpose	How its output was checked
ChatGPT (OpenAI)	Second labelling of the blind scenario pack, used only as a sensitivity check	Compared with the operator's labels; agreement and disagreements reported
Claude (Anthropic), through Claude Code	Coding assistance across all layers, tests, documentation, the Bayesian study, the logo, the system overview figure (Figure 1), the visuals, speaker scripts, audio editing and generated background music of the presentation videos, and drafting help for this report	Every change reviewed and merged by the operator through pull requests with automated tests; results regenerated by scripts in the repository
OpenAI gpt-4.1-mini (API)	A component of the system: specialist agents, the AI Investigator and the post-generation judge	Evaluated as part of the system (Section 8.5); never sets severity or takes action

The presentation videos are narrated by team members; no AI-generated speech is used. All reported numbers come from scripts in the repository and were checked against their outputs.

Acknowledgement of AI use. We used ChatGPT and Claude as tools to assist with research, coding, analysis, drafting and editing. All design decisions, experiments, results and conclusions were reviewed, verified and approved by the team, and we take full responsibility for the content of this report and the submitted system.

Appendix F. References

References follow the IEEE style, numbered in order of first citation.

1. D. Donnellan, A. Lawrence, and R. Weinschenk, "Annual outage analysis 2025," Uptime Institute Intelligence, Keynote Rep. 173, May 2025. [Online]. Available: https://intelligence.uptimeinstitute.com/resource/annual-outage-analysis-2025
2. Data Center Knowledge, "Data center outages decline for fourth straight year, but issues persist," 2025. [Online]. Available: https://www.datacenterknowledge.com/outages/data-center-outages-decline-for-fourth-straight-year-but-issues-persist
3. TechSci Research, "Data center physical security market – Global industry size, share, trends, opportunity, and forecast, 2020–2030F," Research and Markets, Sep. 2025. [Online]. Available: https://www.researchandmarkets.com/reports/6177052/data-center-physical-security-market-global
4. Fortune Business Insights, "Data center physical security market size, share & industry analysis, 2026–2034," Rep. FBI116283, updated Oct. 2026. [Online]. Available: https://www.fortunebusinessinsights.com/data-center-physical-security-market-116283
5. The Insight Partners, "Data center physical security market: Forecast to 2034," 2026. [Online]. Available: https://www.theinsightpartners.com/reports/data-center-physical-security-market
6. B. A. Alahmadi, L. Axon, and I. Martinovic, "99% false positives: A qualitative study of SOC analysts' perspectives on security alarms," in *Proc. 31st USENIX Security Symp.*, Boston, MA, USA, 2022, pp. 2783–2800.
7. F. Valeur, G. Vigna, C. Kruegel, and R. A. Kemmerer, "A comprehensive approach to intrusion detection alert correlation," *IEEE Trans. Dependable Secure Comput.*, vol. 1, no. 3, pp. 146–169, 2004, doi: 10.1109/TDSC.2004.21.
8. A. Hogan *et al.*, "Knowledge graphs," *ACM Comput. Surv.*, vol. 54, no. 4, Art. no. 71, 2021, doi: 10.1145/3447772.

9. L. F. Sikos, "Cybersecurity knowledge graphs," *Knowl. Inf. Syst.*, vol. 65, pp. 3511–3531, 2023, doi: 10.1007/s10115-023-01860-3.
10. K. P. Ahmed, A. Mourin, and K. M. U. Ahmed, "Application of predictive maintenance in Industry 4.0: A use-case study for data centers," in *Proc. Int. Conf. Sci. Technol. Innov. (STI)*, 2021, doi: 10.1109/STI53101.2021.9732590.
11. M. Botezatu, I. Giurgiu, J. Bogojeska, and D. Wiesmann, "Predicting disk replacement towards reliable data centers," in *Proc. 22nd ACM SIGKDD Int. Conf. Knowl. Discov. Data Mining (KDD)*, 2016, doi: 10.1145/2939672.2939699.
12. F. T. Liu, K. M. Ting, and Z.-H. Zhou, "Isolation forest," in *Proc. 8th IEEE Int. Conf. Data Mining (ICDM)*, 2008, pp. 413–422, doi: 10.1109/ICDM.2008.17.
13. M. Sakurada and T. Yairi, "Anomaly detection using autoencoders with nonlinear dimensionality reduction," in *Proc. 2nd Workshop Mach. Learn. Sensory Data Anal. (MLSDA)*, 2014, pp. 4–11, doi: 10.1145/2689746.2689747.
14. J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2016, pp. 779–788, doi: 10.1109/CVPR.2016.91.
15. G. Jocher, A. Chaurasia, and J. Qiu, "Ultralytics YOLOv8," 2023. [Online]. Available: https://github.com/ultralytics/ultralytics
16. J. Deng, J. Guo, E. Ververas, I. Kotsia, and S. Zafeiriou, "RetinaFace: Single-shot multi-level face localisation in the wild," in *Proc. IEEE/CVF CVPR*, 2020.
17. J. Deng, J. Guo, N. Xue, and S. Zafeiriou, "ArcFace: Additive angular margin loss for deep face recognition," in *Proc. IEEE/CVF CVPR*, 2019.
18. J. Pearl, *Probabilistic Reasoning in Intelligent Systems: Networks of Plausible Inference*. San Mateo, CA, USA: Morgan Kaufmann, 1988.
19. P. Lewis *et al.*, "Retrieval-augmented generation for knowledge-intensive NLP tasks," in *Proc. NeurIPS*, 2020. [Online]. Available: https://arxiv.org/abs/2005.11401
20. Z. Ji *et al.*, "Survey of hallucination in natural language generation," *ACM Comput. Surv.*, vol. 55, no. 12, Art. no. 248, 2023, doi: 10.1145/3571730.
21. Schneider Electric, "EcoStruxure IT data center infrastructure management." [Online]. Available: https://www.se.com/ww/en/work/solutions/data-centers-and-networks/
22. Vertiv, "Trellis platform." [Online]. Available: https://www.vertiv.com/en-us/products-catalog/monitoring-control-and-management/software/trellis-platform/
23. IBM, "Maximo asset performance management." [Online]. Available: https://www.ibm.com/products/maximo/asset-performance-management
24. Splunk, "Splunk Enterprise Security." [Online]. Available: https://www.splunk.com/en_us/products/enterprise-security.html
25. Microsoft, "Microsoft Sentinel." [Online]. Available: https://www.microsoft.com/security/business/siem-and-xdr/microsoft-sentinel
26. Genetec, "Security Center unified security platform." [Online]. Available: https://www.genetec.com/products/unified-security/security-center
27. Backblaze, "Hard drive data and stats (Drive Stats)," 2025 Q2–2026 Q1. [Online]. Available: https://www.backblaze.com/cloud-storage/resources/hard-drive-test-data
28. J. Zhu, S. He, P. He, J. Liu, and M. R. Lyu, "Loghub: A large collection of system log datasets for AI-driven log analytics," in *Proc. IEEE Int. Symp. Softw. Rel. Eng. (ISSRE)*, 2023.
29. V. Boiko and O. Niiakyi, "A 4-month dataset of SSH botnet interactions and command payloads," Zenodo, v2.3, 2026, doi: 10.5281/zenodo.20435481. Data paper: *Data in Brief*, vol. 66, Art. no. 112860, 2026.
30. H. M. Ahmad and A. Rahimi, "SH17: A dataset for human safety and personal protective equipment detection in manufacturing industry," *J. Saf. Sci. Resil.*, vol. 6, no. 2, pp. 175–185, 2025, doi: 10.1016/j.jnlssr.2024.09.002.

31. N. B. Naik, "Safety helmet and reflective jacket," Kaggle, 2023. [Online]. Available: https://www.kaggle.com/datasets/niravnaik/safety-helmet-and-reflective-jacket
32. Z. Wang, Y. Wu, L. Yang, A. Thirunavukarasu, C. Evison, and Y. Zhao, "Fast personal protective equipment detection for real construction sites using deep learning approaches," *Sensors*, vol. 21, no. 10, Art. no. 3478, 2021, doi: 10.3390/s21103478.
33. A. Zagorskii, "Face verification dataset," Kaggle, 2025. [Online]. Available: https://www.kaggle.com/datasets/juice0lover/face-identification
34. NIST, "Incident response recommendations and considerations for cybersecurity risk management," SP 800-61 Rev. 3, 2025. [Online]. Available: https://csrc.nist.gov/pubs/sp/800/61/r3/final
35. NIST, "Guide to enterprise patch management planning," SP 800-40 Rev. 4, 2022. [Online]. Available: https://csrc.nist.gov/pubs/sp/800/40/r4/final
36. NIST, "Security and privacy controls for information systems and organizations," SP 800-53 Rev. 5 (Physical and Environmental Protection family). [Online]. Available: https://csrc.nist.gov/pubs/sp/800/53/r5/upd1/final
37. NIST, "Contingency planning guide for federal information systems," SP 800-34 Rev. 1. [Online]. Available: https://csrc.nist.gov/pubs/sp/800/34/r1/upd1/final
38. CISA, "Cybersecurity incident & vulnerability response playbooks." [Online]. Available: https://www.cisa.gov/sites/default/files/publications/Cybersecurity_Incident_Vulnerability_Response_Playbooks_508C.pdf
39. Interagency Security Committee, "Facility access control: An Interagency Security Committee best practice," CISA, 2020. [Online]. Available: https://www.cisa.gov/sites/default/files/2025-02/Facility%20Access%20Control%20-%20An%20Interagency%20Security%20Committee%20Best%20Practice-02-20.pdf
40. OSHA, *29 CFR 1910.132, 1910.135, 1910.137, 1910.147 and 1910.38*. [Online]. Available: https://www.osha.gov/laws-regs/regulations/standardnumber/1910
41. U.S. Dept. of Energy FEMP, "Operations and maintenance best practices"; "O&M challenges and solutions"; "Best practices guide for energy-efficient data center design" (2024); "O&M in federal facilities." [Online]. Available: https://www.energy.gov/cmei/femp
42. DC-GUARDIAN project, "Operational response policy v1," internal draft, 2026.
43. S. I. Serengil and A. Ozpinar, "LightFace: A hybrid deep face recognition framework," in *Proc. Innov. Intell. Syst. Appl. Conf. (ASYU)*, 2020.
44. N. Reimers and I. Gurevych, "Sentence-BERT: Sentence embeddings using Siamese BERT-networks," in *Proc. EMNLP*, 2019.
45. W. Wang, F. Wei, L. Dong, H. Bao, N. Yang, and M. Zhou, "MiniLM: Deep self-attention distillation for task-agnostic compression of pre-trained transformers," in *Proc. NeurIPS*, 2020.
46. J. Brooke, "SUS: A 'quick and dirty' usability scale," in Usability Evaluation in Industry, P. W. Jordan, B. Thomas, B. A. Weerdmeester, and I. L. McClelland, Eds. London, U.K.: Taylor & Francis, 1996, pp. 189–194.
47. A. Bangor, P. Kortum, and J. Miller, "Determining what individual SUS scores mean: Adding an adjective rating scale," J. Usability Stud., vol. 4, no. 3, pp. 114–123, 2009.

Abbreviations

Abbreviation	Meaning
AI	artificial intelligence
API	application programming interface
BMS	building management system
CI	continuous integration
CISA	Cybersecurity and Infrastructure Security Agency (U.S.)
CPU	central processing unit

Abbreviation	Meaning
CSV	comma-separated values
DCIM	data center infrastructure management
DOE	Department of Energy (U.S.)
FEMP	Federal Energy Management Program (U.S. DOE)
GPU	graphics processing unit
GRU	gated recurrent unit
IoT	Internet of Things
IP	Internet Protocol (address)
JSON	JavaScript Object Notation
LFS	Large File Storage (Git LFS)
LLM	large language model
mAP	mean average precision
MRR	mean reciprocal rank
NIST	National Institute of Standards and Technology (U.S.)
OSHA	Occupational Safety and Health Administration (U.S.)
PPE	personal protective equipment
PR-AUC	area under the precision–recall curve
RAG	retrieval-augmented generation
ROC-AUC	area under the receiver operating characteristic curve
SHA-256	Secure Hash Algorithm, 256-bit
SIEM	security information and event management
SMART	Self-Monitoring, Analysis and Reporting Technology (drives)
SOC	security operations center
SSH	Secure Shell
SUS	System Usability Scale
TLS	Transport Layer Security
UI	user interface
YOLO	You Only Look Once (object detector)

Acknowledgements

We thank the NUS-ISS lecturers of the Intelligent Reasoning Systems certificate for their guidance, Saudi Aramco Company for sponsoring us to study the Master of Technology in Artificial Intelligence Systems, the participants of our usability study for their time, and the authors of the public datasets and open-source tools listed in Appendix F, without which this project would not have been possible.